

Advanced Monograph Series

Theory Of Cauchy Equations

A systematic exposition of the theory of Cauchy
functional equations

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Theory of Cauchy Equations

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This monograph is intended as a rigorous analytical exposition of the theory of Cauchy functional equations and related structures, including additive and multiplicative equations, regularity conditions, continuous solutions, differentiable solutions, analytical solutions, transformable solutions and associated developments in functional analysis.

The author does not claim original discovery of the mathematical results, theorems, or constructions presented in this work. This text is a systematic and rigorous exposition of established mathematical theory, prepared with the aim of clarity, accessibility, and deeper analytical understanding. All primary references and source materials consulted during the preparation of this work are listed in the references section.

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Preface

I wrote this book because I found the theory of Cauchy equations extraordinarily interesting and wanted a single rigorous source where the subject could be studied together instead of through scattered and unrelated chapters across different materials.

Most people see the basic Cauchy equation once and move on, but that should not be the case. Behind its simple appearance lies a surprisingly deep and beautiful theory connected to analysis, algebra, topology, and the foundations of mathematics itself. Some of the strangest and most fascinating objects in mathematics emerge naturally from these equations.

This book is an attempt to present that theory carefully, rigorously, and in a readable form for anyone willing to explore the subject seriously.

I have prepared this with patience, curiosity and unbelievable amounts of hand-written calculations.

Sharad Anirudh Jonnalagadda

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Solving The Famous Cauchy Equations

We will be discussing continuous Cauchy functional equations to start things off as these results will serve as building blocks for all the upcoming sections. There are four such designated functional equations attributed to Louis Augustine Cauchy and we will be studying them rigorously in the upcoming expositions. Firstly, let us introduce ourselves to the Linear Cauchy functional equation.

Let, $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x, y \in \mathbb{R}$ satisfies the given functional equation:

$$f(x + y) = f(x) + f(y)$$

Our task is to find all such applicable f that satisfy the given functional equation.

To begin the exposition, assume $y = z + w$ such that $z, w \in \mathbb{R}$ due to which the we obtain the following identity:

$$f(x + z + w) = f(x) + f(z + w) = f(x) + f(z) + f(w)$$

Similarly, if we assume $w = u + v$ such that $u, v \in \mathbb{R}$, we obtain:

$$\begin{aligned} f(x + z + u + v) &= f(x) + f(z + u + v) = f(x) + f(z) + f(u + v) \\ &= f(x) + f(z) + f(u) + f(v) \end{aligned}$$

This implies, for some $k \in \mathbb{N}$ and $k \geq 2$, we can rewrite the functional equation as such:

$$f\left(\sum_{1 \leq i \leq k} x_i\right) = \sum_{1 \leq i \leq k} f(x_i)$$

Such that, $x_i \in \mathbb{R}$ for every $1 \leq i \leq k$. The above can be proved by using mathematical induction by setting the base case as $k = 2$ and then proving the generalized case.

Now, we set $x_i = x$ for every $1 \leq i \leq k$, by which we have:

$$f(kx) = kf(x)$$

Now here comes the most important step in this derivation, we say $x = \frac{mt}{k}$ such that $m \in \mathbb{N}, t \in \mathbb{R}$. Because of which, we have:

$$f(mt) = kf\left(\frac{mt}{k}\right)$$

By the linearity property of the functional equation, we can readily state that $f(mt) = mf(t)$ as $m \in \mathbb{N}$. By using this and dividing k on both sides (this step is justified as $k \in \mathbb{N}$ and cannot be zero) we have:

$$f\left(\frac{mt}{k}\right) = \frac{m}{k}f(t)$$

As both $m, k \in \mathbb{N}$, we can say $q = \frac{mt}{k}$ such that $q \in \mathbb{Q}$.

$$f(qt) = qf(t)$$

The introduction of a rational value q was important as it lets us set a sequence of rational numbers denoted by $\{q\}$ such that its limiting behaviour can be defined as follows:

$$\lim_{k \rightarrow \infty} (q_k) = \mu$$

Such that $\mu \in \mathbb{R}$. Which ultimately means, we can rewrite the second last previous expression as such:

$$f(q_k t) = q_k f(t)$$

Since, f is a continuous function we can confidently state the following:

$$\lim_{n \rightarrow \infty} f(q_k t) = f\left(\lim_{k \rightarrow \infty} q_k t\right) = f(\mu t) = \mu f(t)$$

Substitute $t = 1$ in the above expression and here we assume that $f(1) = a$ such that $a \in \mathbb{R}$.

$$f(\mu) = a\mu$$

The above is our final solution. Needless to say, one can substitute $\mu = x$ or $\mu = y$ to get the result in terms $f(x)$ or $f(y)$. With that done, we now explore the next Cauchy equation also called as exponential Cauchy equation. The problem statement is as follows:

Let, $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x, y \in \mathbb{R}$ the following functional equation is satisfied.

$$f(x + y) = f(x)f(y)$$

To find all such applicable f .

Once again, we can generalize the given functional equation as follows:

$$f\left(\sum_{1 \leq i \leq k} x_i\right) = \prod_{1 \leq i \leq k} f(x_i)$$

Such that $k \in \mathbb{N}$, $k \geq 2$ and $x_i \in \mathbb{R}$ for every $1 \leq i \leq k$. The above can be proved using mathematical induction by setting the base case as $k = 2$ and then proving the generalized case.

Now, we set $x_i = x$ for every $1 \leq i \leq k$, by which we have:

$$f(kx) = f(x)^k$$

Here, we say $x = \frac{mt}{k}$ such that $m \in \mathbb{N}$, $t \in \mathbb{R}$. Because of which we have:

$$f(mt) = f\left(\frac{mt}{k}\right)^k$$

Now, by using the previously established power identity and by scaling the powers in above expression, we can state the following:

$$f(t)^m = f\left(\frac{mt}{k}\right)^k \rightarrow f(t)^{\frac{m}{k}} = f\left(\frac{mt}{k}\right)$$

Equivalent to our previous exposition, we say $q = \frac{m}{k}$ such that $q \in \mathbb{Q}$.

$$f(qt) = f(t)^q$$

Assuming a series of rational numbers say $\{q\}$ having the following limiting behaviour:

$$\lim_{k \rightarrow \infty} (q_k) = \mu$$

Such that $\mu \in \mathbb{R}$. Which ultimately implies the following:

$$f(q_k t) = f(t)^{q_k}$$

As we already know that f is a continuous function, we can confidently state the following:

$$\lim_{k \rightarrow \infty} f(q_k t) = f\left(\lim_{k \rightarrow \infty} q_k t\right) = f(\mu t) = f(t)^\mu$$

We substitute $t = 1$ and we also assume that $f(1) = b$ such that $b \in \mathbb{R}$, because of which our final solution is:

$$f(\mu) = b^\mu$$

The same can be verified by substituting the solution back into the original functional equation. Let us now explore the next Cauchy equation, popularly known as the logarithmic Cauchy equation. The problem statement is as follows:

Let, $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that f is a continuous function and $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(xy) = f(x) + f(y)$$

Find all such applicable f .

Similar to our previous expositions, we can write the generalized version of this functional equation as follows:

$$f\left(\prod_{1 \leq i \leq k} x_i\right) = \sum_{1 \leq i \leq k} f(x_i)$$

Such that $k \in \mathbb{N}, k \geq 2$ and $x_i \in \mathbb{R}^+$ for every $1 \leq i \leq k$. We substitute $x_i = x$ for every $1 \leq i \leq k$.

$$f(x^k) = kf(x)$$

Here, we say $x = t^{\frac{m}{k}}$ such that $t \in \mathbb{R}^+$ and $m \in \mathbb{N}$. By which we have:

$$f(t^m) = mf(t) = kf\left(t^{\frac{m}{k}}\right)$$

On rearranging the terms, we obtain the following identity:

$$f\left(t^{\frac{m}{k}}\right) = \frac{m}{k}f(t)$$

As per usual, we say $q = \frac{m}{k}$ such that $q \in \mathbb{Q}$.

$$f(t^q) = qf(t)$$

Assuming a sequence of rational numbers $\{q\}$ having the following limiting behaviour:

$$\lim_{k \rightarrow \infty} \{q_k\} = \mu$$

Which instantly allows us to rewrite the stated identity:

$$f(t^{q_k}) = q_k f(t)$$

As we know that f is a continuous function, we can state the following:

$$\lim_{k \rightarrow \infty} f(t^{q_k}) = f\left(\lim_{k \rightarrow \infty} t^{q_k}\right) = f(t^\mu) = \mu f(t)$$

Now, we say $t = b$ such that b is some constant and $b \in \mathbb{R}^+$ and $b^\mu = A$ as $A \in \mathbb{R}^+$ such that $\mu = \log_b A$, which enables us to state the following final solution:

$$f(A) = \log_b A(f(b)) = \log_{b^{\frac{1}{f(b)}}} A = \log_c A$$

Here, $c = b^{\frac{1}{f(b)}}$ as $c \in \mathbb{R}^+$. Finally let us solve the last Cauchy's functional equation also known as multiplicative Cauchy equation. The problem statement is given below:

Let, $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(xy) = f(x)f(y)$$

To find all such applicable f .

As always, generalizing the given functional equation:

$$f\left(\prod_{1 \leq i \leq k} x_i\right) = \prod_{1 \leq i \leq k} f(x_i)$$

Such that $k \in \mathbb{N}_{\geq 2}$ and $x_i \in \mathbb{R}^+$ for every $1 \leq i \leq k$. Here, we set $x_i = x$ for every $1 \leq i \leq k$, by which we have:

$$f(x^k) = f(x)^k$$

As per usual, we say $x = t^{\frac{m}{k}}$ such that $t \in \mathbb{R}^+$ and $m \in \mathbb{N}$. Because of which, we have:

$$f(t^m) = f(t)^m = f\left(t^{\frac{m}{k}}\right)^k$$

On rearranging the terms and scaling the powers on both sides, we obtain the following:

$$f\left(t^{\frac{m}{k}}\right) = f(t)^{\frac{m}{k}}$$

Now, we say $q = \frac{m}{k}$ such that $q \in \mathbb{Q}$. By which we have:

$$f(t^q) = f(t)^q$$

We assume a sequence of rational numbers $\{q\}$ exist such that the following identity holds:

$$\lim_{k \rightarrow \infty} q_k = \mu$$

Such that $\mu \in \mathbb{R}^+$, As f is continuous, we can confidently state the following:

$$\lim_{k \rightarrow \infty} f(t^{q_k}) = f\left(\lim_{k \rightarrow \infty} t^{q_k}\right) = f(t^\mu) = f(t)^\mu$$

Let, $t = a$ such that $a \in \mathbb{R}^+$, $a^\mu = b$ such that $b \in \mathbb{R}^+$ and $f(a) = c$ assuming $c \in \mathbb{R}^+$. By making use of all this, we have:

$$f(b) = c^\mu = c^{\log b / \log a} = (c^{\log_c b})^{\log_a c} = b^{\log_a c}$$

Here, we can assume $p = \log_a c$ such that $p \in \mathbb{R}$. Thus, our final solution being:

$$f(b) = b^p$$

With these four fundamental functional equations solved, it's time to explore few examples which are all Cauchy equations in disguise.

Example 1.1: Let h be a continuous function mapped from $h : \mathbb{R} \rightarrow \mathbb{R}$ and $\forall x, y, c \in \mathbb{R}$, the following functional equation remains satisfied:

$$h(x + y) = h(x) + h(y) + c$$

To find all such applicable h .

Realize that the original equation can be rewritten as follows:

$$h(x + y) + c = (h(x) + c) + (h(y) + c)$$

Now let us assume, $H(x) = h(x) + c$ such that $H : \mathbb{R} \rightarrow \mathbb{R}$. Using which, we have:

$$H(x + y) = H(x) + H(y)$$

This is simply Linear Cauchy equation we had solved earlier.

$$H(x) = ax \rightarrow h(x) = ax - c$$

Here, $a \in \mathbb{R}$ and that is our final solution.

Example 1.2: Let g be a continuous function mapped from $g : \mathbb{R} \rightarrow \mathbb{R}$ and $\forall x, y, b \in \mathbb{R}$ the following functional equation remains satisfied:

$$g(x + y) = g(x + b) + g(y - b)$$

To find all such applicable g .

We rewrite the original functional equation as follows:

$$g(x + b + y - b) = g(x + b) + g(y - b)$$

Now, let us assume that $X = x + b$ and $Y = y - b$. Needless to say, $X, Y \in \mathbb{R}$ because of which we have:

$$g(X + Y) = g(X) + g(Y)$$

Once again, we have obtained a Linear Cauchy functional equation.

$$g(X) = aX \rightarrow g(x) = ax$$

Here, $a \in \mathbb{R}$ and above is our final solution.

Example 1.3: Let f be a continuous function mapped from $f : \mathbb{R} \rightarrow \mathbb{R}$ and $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$f(x + y) = e^{-y}f(x) + e^{-x}f(y)$$

To find all such applicable f .

We multiply the term e^{x+y} on both sides of the given functional equation to obtain the following:

$$e^{x+y}f(x + y) = e^x f(x) + e^y f(y)$$

Here, we assume $F(x) = e^x f(x)$, because of which, we obtain Linear Cauchy equation yet again.

$$F(x + y) = F(x) + F(y)$$

The final solution being:

$$F(x) = ax \rightarrow f(x) = axe^x$$

Here, $a \in \mathbb{R}$.

Example 1.4: Let F be a continuous function mapped from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $k \in \mathbb{N}$ the following functional equation is satisfied.

$$F\left(\frac{x+y}{k}\right) = \frac{F(x) + F(y)}{k}$$

To find all such applicable F .

Let us set $y = 0$ to start things off.

$$F\left(\frac{x}{k}\right) = \frac{F(x)}{k}$$

Now, let us say $x = u + v$ such that $u, v \in \mathbb{R}$, by which we have:

$$F\left(\frac{u+v}{k}\right) = \frac{F(u+v)}{k}$$

But we also do know from the given functional equation, that the following identity must also satisfy:

$$F\left(\frac{u+v}{k}\right) = \frac{F(u) + F(v)}{k}$$

By comparing the two, we obtain yet another Linear Cauchy functional equation.

$$F(u) + F(v) = F(u+v)$$

Thus, our final solution is given as:

$$F(x) = ax$$

As per usual, $a \in \mathbb{R}$.

Example 1.5: Let g be continuous function mapped from $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $A \in \mathbb{N}$ the following functional equation is satisfied.

$$g(x+y) = A^{2xy} g(x)g(y)$$

Find all such applicable g .

The original functional equation can be rewritten as:

$$A^{-(x+y)^2} g(x+y) = \left(A^{-x^2} g(x)\right) \left(A^{-y^2} g(y)\right)$$

Here, we set $G(x) = A^{-x^2} g(x)$, Because of which we can rewrite the original functional equation as such:

$$G(x + y) = G(x)G(y)$$

This is simply exponential Cauchy equation and its solution is stated down below:

$$G(x) = b^x \Rightarrow g(x) = A^{-x^2} b^x$$

Here, $b \in \mathbb{R}$.

Example 1.6: Let h be a continuous function mapped from $h : \mathbb{R} \rightarrow \mathbb{R}$ and $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$h(x + y) = h(x) + h(y) + h(x)h(y)$$

To find all such applicable h .

Notice that the original functional equation can be factorized.

$$1 + h(x + y) = (1 + h(x))(1 + h(y))$$

Set $H(x) = 1 + h(x)$, by which we have:

$$H(x + y) = H(x)H(y)$$

This is yet again an exponential Cauchy equation and its solution being:

$$H(x) = b^x \Rightarrow h(x) = b^x - 1$$

Here, $b \in \mathbb{R}$.

Let us now proceed to the next section where we will discuss generalized Cauchy equations in detail:

Generalized Cauchy Equations

Generalized Cauchy equations also known as Pexider's equations are evaluations of functional equations with $n + 1$ such functionals. Let us discuss the first Pexider's equation. To start things off, let us start with a simple example, and scaffold from thereon.

Let, f, g, h are continuous functions mapped from $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation is satisfied.

$$f(x + y) = g(x) + h(y)$$

To find all such applicable f, g, h .

Firstly, set $x = 0$ and then $y = 0$ to obtain the following system of equations:

$$f(y) = g(0) + h(y)$$

$$f(x) = g(x) + h(0)$$

Adding these equations and then making use of the definition itself gives us the following:

$$f(x) + f(y) = (g(x) + h(y)) + (g(0) + h(0))$$

$$f(x) + f(y) = f(x + y) + f(0)$$

Now, set $F(x) = f(x) - f(0)$, resulting in the following Linear Cauchy equation.

$$F(x) + F(y) = F(x + y)$$

Therefore, the solution to the above equation is as follows (Here, we assume $a, b \in \mathbb{R}$ and $f(0) = b$).

$$F(x) = ax \Rightarrow f(x) = ax + b$$

Specific solutions for both g, h can be calculated quite easily by using the earlier defined system of equations:

$$g(x) = ax + c, h(x) = ax + d$$

We obtain the above by considering $b - g(0) = c, b - h(0) = d$ such that $c, d \in \mathbb{R}$ and $b = c + d$.

Now, considering the generalized case. Let, f, f_i be continuous functions such that $f, f_i : \mathbb{R} \rightarrow \mathbb{R}$ and for every $x_i \in \mathbb{R}$ assuming $1 \leq i \leq n$, the following functional equation remains satisfied (Note: Here, it is assumed that $n > 2$).

$$f\left(\sum_{1 \leq i \leq n} x_i\right) = \sum_{1 \leq i \leq n} f_i(x_i)$$

By setting $x_j = 0$ for every $1 \leq j \leq n$ such that $j \neq i$, we obtain a system of n equations which can be described aptly as follows:

$$f(x_i) = f_i(x_i) + \sum_{\substack{1 \leq j \leq n \\ j \neq i}} f_j(0)$$

On adding all these equations, we obtain the following:

$$\sum_{1 \leq i \leq n} f(x_i) = f\left(\sum_{1 \leq i \leq n} x_i\right) + (n-1)f(0)$$

Here, we say $F(x_i) = f(x_i) - f(0)$, to reduce the above expression.

$$\sum_{1 \leq i \leq n} F(x_i) = F\left(\sum_{1 \leq i \leq n} x_i\right)$$

The above expression is simply the extrapolated version of the linear Cauchy equation, because of which the general solution of F is (Note: Assume $x \in \mathbb{R}$):

$$F(x) = ax \Rightarrow f(x) = ax + b$$

Here, $a, b \in \mathbb{R}$, $x \in \mathbb{R}$ and $f(0) = b$. Using this general result, it becomes almost trivial to determine all f_i as well.

$$f_i(x) = ax + b_i$$

Such that every $b_i \in \mathbb{R}$ (Where, $1 \leq i \leq n$) and $\sum_{1 \leq i \leq n} b_i = b$.

Similarly, let us now work with second generalized Cauchy equation but before that, we solve a simpler version to understand its mechanics properly. The problem statement given as follows:

Let, f, g, h are continuous functions mapped from $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation is satisfied.

$$f(x+y) = g(x)h(y)$$

To find all such applicable f, g, h .

We set $x = 0$ and then $y = 0$ to obtain the following system of equations:

$$f(x) = g(x)h(0)$$

$$f(y) = g(0)h(y)$$

We multiply these two to obtain the following resulting expression:

$$f(x)f(y) = (g(x)h(y))(g(0)h(0))$$

By making use of the definition of the given functional equation we can formulate the following (Note: It is assumed that $f(0) \neq 0$):

$$f(x)f(y) = f(x+y)f(0)$$

The above functional equation can be rewritten as follows:

$$\frac{f(x)}{f(0)} \cdot \frac{f(y)}{f(0)} = \frac{f(x+y)}{f(0)}$$

By setting $F(x) = \frac{f(x)}{f(0)}$, we have:

$$F(x+y) = F(x)F(y)$$

Which is simply exponential Cauchy equation, with its solution being (Assuming, $b \in \mathbb{R}$, $a \in \mathbb{R} / \{0\}$ and $f(0) = \frac{1}{a}$):

$$F(x) = b^x \rightarrow f(x) = ab^x$$

Thus, solutions of g, h are given as:

$$g(x) = cb^x, h(x) = db^x$$

Such that $a/h(0) = c$, $a/g(0) = d$ which implies $a = cd$ where $c, d \in \mathbb{R}$. With that done let us discuss the generalized case.

Let, f, f_i be continuous functions such that $f, f_i : \mathbb{R} \rightarrow \mathbb{R}$ and for every $x_i \in \mathbb{R}$ assuming $1 \leq i \leq n$, the following functional equation remains satisfied (Note: Here, it is assumed that $n > 2$).

$$f\left(\sum_{1 \leq i \leq n} x_i\right) = \prod_{1 \leq i \leq n} f_i(x_i)$$

Now, we set $x_j = 0$ procedurally for every $1 \leq j \leq n$ and $j \neq i$ to obtain a system of equations which can be aptly represented by the following notation:

$$f(x_i) = f_i(x_i) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} f_j(0)$$

Once we multiply all the n equations and use the definition of the functional equation, we obtain the following:

$$\begin{aligned} \prod_{1 \leq i \leq n} f(x_i) &= \prod_{1 \leq i \leq n} f_i(x_i) (f(0))^{n-1} \\ \prod_{1 \leq i \leq n} f(x_i) &= f\left(\sum_{1 \leq i \leq n} x_i\right) (f(0))^{n-1} \end{aligned}$$

On rearranging the terms, we obtain the following:

$$\prod_{1 \leq i \leq n} \frac{f(x_i)}{f(0)} = \frac{f(\sum_{1 \leq i \leq n} x_i)}{f(0)}$$

Here, we say $F(x_i) = \frac{f(x_i)}{f(0)}$, by which we obtain an exponential Cauchy equation.

$$\prod_{1 \leq i \leq n} F(x_i) = F\left(\sum_{1 \leq i \leq n} x_i\right)$$

Observe that this is the generalized version of the exponential Cauchy equation and as a virtue, the following will be its general solution assuming $f(0) = \frac{1}{a}$ such that $a \in \mathbb{R} / \{0\}$ and $b, x \in \mathbb{R}$.

$$F(x) = b^x \Rightarrow f(x) = ab^x$$

Similarly, a general solution for f_i can be expressed as follows assuming $a_i \in \mathbb{R}$ and $\prod_{i=1}^n a_i = a$.

$$f_i(x) = a_i b^x$$

Let us now look at the generalized logarithmic Cauchy equation but as per usual, let us work out a simpler problem first before diving into the generalization.

Let, f, g, h be continuous functions defined from $f, g, h : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(xy) = g(x) + h(y)$$

To find all such applicable f, g, h .

Firstly, setting $y = 1$ and then setting $x = 1$ to obtain the following system of equations.

$$f(x) = g(x) + h(1)$$

$$f(y) = g(1) + h(y)$$

Adding the resultant equations and making use of the definition gives us the following:

$$f(x) + f(y) = (g(x) + h(y)) + (g(1) + h(1)) = f(xy) + f(1)$$

On rearranging the terms, we have:

$$(f(x) - f(1)) + (f(y) - f(1)) = (f(xy) - f(1))$$

Now, we say $F(x) = f(x) - f(1)$, by which we have:

$$F(x) + F(y) = F(xy)$$

Do notice that this is the logarithmic Cauchy equation and its general solution is written down as follows assuming $a, c \in \mathbb{R}^+$ and $f(1) = a$.

$$F(x) = \log_c x \rightarrow f(x) = \log_c x + a$$

Similarly, general solutions of g, h are given as follows:

$$g(x) = \log_c x + b, h(x) = \log_c x + d$$

Such that $b, d \in \mathbb{R}^+$ and $a = b + d$.

With that done, we can now discuss the general variant of this problem.

Let, f, f_i be continuous functions defined from $f, f_i : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x_i \in \mathbb{R}^+$ given $1 \leq i \leq n$ the following functional equation remains satisfied.

$$f\left(\prod_{1 \leq i \leq n} x_i\right) = \sum_{1 \leq i \leq n} f_i(x_i)$$

To find all such applicable f, f_i .

To start things off, we methodically set $x_j = 1$ for every $1 \leq j \leq n$ and $i \neq j$ obtaining the following:

$$f(x_i) = f_i(x_i) + \sum_{\substack{1 \leq j \leq n \\ j \neq i}} f_j(1)$$

Now, we will add up all the resulting n equations and make use of the definition.

$$\sum_{1 \leq i \leq n} f(x_i) = \sum_{1 \leq i \leq n} f_i(x_i) + (n-1)f(1) = f\left(\prod_{1 \leq i \leq n} x_i\right) + (n-1)f(1)$$

On rearranging the terms, we have:

$$\sum_{1 \leq i \leq n} (f(x_i) - f(1)) = f\left(\prod_{1 \leq i \leq n} x_i\right) - f(1)$$

Here, we say $F(x_i) = f(x_i) - f(1)$ because of which the above functional equation reduces to the following:

$$\sum_{1 \leq i \leq n} F(x_i) = F\left(\prod_{1 \leq i \leq n} x_i\right)$$

Do notice that this is logarithmic Cauchy equation in n variables. Solution to which is given as follows assuming $a, x, c \in \mathbb{R}^+$ and $f(1) = a$.

$$F(x) = \log_c x \Rightarrow f(x_i) = \log_c x + a$$

Similarly, we can confidently state the general solution of f_i assuming $\sum_{1 \leq i \leq n} a_i = a$ and $a_i \in \mathbb{R}^+$.

$$f_i(x) = \log_c x + a_i$$

With that done, we can now discuss the generalized multiplicative Cauchy equation but as usual, let us discuss a simpler case first.

Let, f, g, h be continuous functions defined from $f, g, h : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(xy) = g(x)h(y)$$

To find all such applicable f, g, h .

Let us set $x = 1$ and then $y = 1$ to obtain the following system of equations;

$$f(x) = g(x)h(1)$$

$$f(y) = g(1)h(y)$$

Multiplying the resulting equations and using the definition results in the following:

$$f(x)f(y) = (g(x)h(y))(g(1)h(1)) = f(xy)f(1)$$

Now, we rearrange the terms (Needless to say, from the conditions we have $f(1) \neq 0$)

$$\frac{f(x)}{f(1)} \cdot \frac{f(y)}{f(1)} = \frac{f(xy)}{f(1)}$$

Here, we say $F(x) = \frac{f(x)}{f(1)}$, by which we have:

$$F(xy) = F(x)F(y)$$

This is simply the multiplicative Cauchy equation which has the following general solution assuming $a, c \in \mathbb{R}^+$ and $f(1) = \frac{1}{a}$:

$$F(x) = x^c \rightarrow f(x) = ax^c$$

Similarly, general solutions for g, h are given as follows assuming $b, d \in \mathbb{R}^+$ and $a = bd$.

$$g(x) = bx^c, h(x) = dx^c$$

Let us now discuss the general variant of this problem.

Let, f, f_i be continuous functions defined from $f, f_i : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x_i \in \mathbb{R}^+$ given $1 \leq i \leq n$ the following functional equation remains satisfied.

$$f\left(\prod_{1 \leq i \leq n} x_i\right) = \prod_{1 \leq i \leq n} f_i(x_i)$$

To find all such applicable f, f_i .

To start things off, we methodically set $x_j = 1$ for every $1 \leq j \leq n$ and $i \neq j$ resulting in the following:

$$f(x_i) = f_i(x_i) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} f_j(1)$$

On multiplying all the resulting n equations and using the definition, we obtain the following:

$$\prod_{1 \leq i \leq n} f(x_i) = f(0)^{n-1} \prod_{1 \leq i \leq n} f_i(x_i) = f(0)^{n-1} f\left(\prod_{1 \leq i \leq n} x_i\right)$$

We now rearrange the terms to simplify further.

$$\prod_{1 \leq i \leq n} \frac{f(x_i)}{f(0)} = \frac{f(\prod_{1 \leq i \leq n} x_i)}{f(0)}$$

Setting $F(x_i) = \frac{f(x_i)}{f(0)}$ results in the following:

$$\prod_{1 \leq i \leq n} F(x_i) = F\left(\prod_{1 \leq i \leq n} x_i\right)$$

Do realize that this is a generalized version of multiplicative Cauchy equation which has the following general solution assuming $a, x \in \mathbb{R}^+, c \in \mathbb{R}$ and $f(1) = \frac{1}{a}$:

$$F(x) = x^c \rightarrow f(x) = ax^c$$

Similarly, general solution of f_i can be stated as such assuming $a_i \in \mathbb{R}^+$ and $a = \prod_{1 \leq i \leq n} a_i$:

$$f_i(x) = a_i x^c$$

Now that we have solved all of the generalized Cauchy equations, let us solve some examples which involve computing general solutions of functional equations that are generalized Cauchy equations in disguise.

Example 2.1: Let, f, g, h be continuous functions mapped from $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $B \in \mathbb{N}$ the following functional equation is satisfied.

$$f(x + y) = B^y g(x) + B^x h(y)$$

To find all such applicable f, g, h .

Realize that the original functional equation can be rewritten as follows:

$$B^{-x-y} f(x + y) = B^{-x} g(x) + B^{-y} h(y)$$

Here, we assume $F(x) = B^{-x} f(x)$, $G(x) = B^{-x} g(x)$ and $H(x) = B^{-x} h(x)$ such that $F, G, H : \mathbb{R} \rightarrow \mathbb{R}$. Because of which we have:

$$F(x + y) = G(x) + H(y)$$

This is simplified linear Pexider equation and its general solution is given as follows assuming $a, b \in \mathbb{R}$ and $F(0) = f(0) = b$.

$$F(x) = ax + b \rightarrow f(x) = B^{-x}(ax + b)$$

Similarly, solutions of g, h can be given as follows assuming $c, d \in \mathbb{R}$ and $b = c + d$:

$$G(x) = ax + c \rightarrow g(x) = B^{-x}(ax + c)$$

$$H(x) = ax + d \Rightarrow h(x) = B^{-x}(ax + d)$$

Example 2.2: Let, P, Q, R be continuous functions mapped from $P, Q, R : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation is satisfied.

$$P(x + y) = 1 + Q(x) + R(y) + Q(x)R(y)$$

Find all such applicable P, Q, R .

On factorizing the right-hand side, we have the following:

$$P(x + y) = (1 + Q(x))(1 + R(y))$$

Here, we assume $F(x) = 1 + Q(x)$ and $G(x) = 1 + R(x)$, because of which we have:

$$P(x + y) = F(x)H(y)$$

Do notice that this Generalized exponential Cauchy equation (exponential Pexider's equation) and its solution is given as follows assuming $a, b \in \mathbb{R}$:

$$P(x) = ab^x$$

Similarly, general solutions of Q, R are given as follows assuming $c, d \in \mathbb{R}$ and $a = cd$:

$$F(x) = cb^x \Rightarrow Q(x) = cb^x - 1$$

$$H(x) = db^x \Rightarrow R(x) = db^x - 1$$

Example 2.3: Let, f, g, h be continuous functions mapped from $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation is satisfied.

$$f(x + y) = g(x) + h(y) + 2xy$$

Find all such applicable f, g, h .

We can rewrite the given functional equation as such:

$$f(x + y) - (x + y)^2 = g(x) + h(y) - x^2 - y^2$$

Here, we assume that $F(x) = f(x) - x^2$, $G(x) = g(x) - x^2$ and $H(x) = h(x) - x^2$, by which we have:

$$F(x + y) = G(x) + H(y)$$

This is generalized linear Cauchy equation (linear Pexider equation) and its general solution is given as follows assuming $a, b \in \mathbb{R}$:

$$F(x) = ax + b \rightarrow f(x) = x^2 + ax + b$$

Similarly, g, h have the following general solution assuming $c, d \in \mathbb{R}$ and $b = c + d$.

$$G(x) = ax + c \rightarrow g(x) = x^2 + ax + c$$

$$H(x) = ax + d \rightarrow h(x) = x^2 + ax + d$$

This completes the segment of generalized Cauchy equations and before exploring vector representations of the Cauchy equations, let us pivot to particular solutions and explore the calculus behind the machinery.

Differentiable Cauchy equations

From the beginning of this book, we have assumed the participating functions to be continuous but, in this segment, we will begin to assume that the participating functions are differentiable in nature. Let us firstly explore Linear Cauchy equations with the differentiability constraint.

Let, f be a differentiable function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$f(x + y) = f(x) + f(y)$$

Find all such applicable f .

As it is provided that f is a differentiable function, we can proceed forward and take the derivative w.r.t either x or y . Let us take derivative w.r.t x on both sides.

$$\frac{d}{dx}(f(x + y)) = \frac{d}{dx}(f(x) + f(y)) \rightarrow f'(x + y) = f'(x)$$

Substituting $x = 0$ in the above expression results in the following assuming $f'(0) = a, f(0) = b$ such that $a, b \in \mathbb{R}$.

$$f'(y) = a \rightarrow f(y) = ay + b = ay$$

Do notice that $f(0) = b = 0$ from the definition itself, and because of which we have the given final answer.

Next up, we will explore the exponential Cauchy equation.

Let, f be a differentiable function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$f(x + y) = f(x)f(y)$$

To find all such applicable f .

Firstly, taking the derivative w.r.t x on both sides.

$$\frac{d}{dx}(f(x + y)) = \frac{d}{dx}(f(x)f(y)) \Rightarrow f'(x + y) = f'(x)f(y)$$

Substituting $x = 0$ on both sides results in the following assuming $f'(0) = a$ such that $a \in \mathbb{R}$.

$$\frac{df(y)}{dy} = af(y) \Rightarrow \frac{df(y)}{f(y)} = a dy$$

Integrating on both sides and ignoring the integration constant results in the following assuming $b = e^a$:

$$\ln(f(y)) = ay \Rightarrow f(y) = e^{ay} = b^y$$

Moving ahead, let us now proceed to solve the logarithmic Cauchy equation with the differentiability constraint.

Let, f be a differentiable function defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied:

$$f(xy) = f(x) + f(y)$$

Find all such applicable f .

Once again, taking a derivative on both sides w.r.t x .

$$\frac{d}{dx}(f(xy)) = \frac{d}{dx}(f(x) + f(y)) \Rightarrow yf'(xy) = f'(x)$$

Substituting $x = 1$ and assuming $f'(1) = a$ such that $a \in \mathbb{R}^+$ results in the following:

$$yf'(y) = a \Rightarrow f'(y) = \frac{a}{y}$$

Integrating on both sides w.r.t y and ignoring the integration constant assuming $e^{\frac{1}{a}} = c$ results in the following:

$$f(y) = a \log_e y = \log_c y$$

Finally, solving the multiplicative Cauchy equation with the differentiability constraint.

Let, f be a differentiable function defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied:

$$f(xy) = f(x)f(y)$$

To find all such applicable f .

As per usual, taking derivative on both sides w.r.t x .

$$\frac{d}{dx}(f(xy)) = \frac{d}{dx}(f(x)f(y)) \Rightarrow yf'(xy) = f'(x)f(y)$$

Substituting $x = 1$ and assuming $f'(1) = a$ such that $a \in \mathbb{R}$ results in the following:

$$yf'(y) = af(y) \Rightarrow \frac{df(y)}{f(y)} = a \frac{dy}{y}$$

Integrating on both sides and ignoring the integration constant results in the following:

$$\log f(y) = a \log y = \log(y^a)$$

$$f(y) = y^a$$

With this done, let us solve few examples based on the learned differentiability constraint.

Example 3.1: Let, F be a differentiable function defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$F(x + y) = F(x) + F(y) + F(x)F(y)$$

Find all such applicable F .

Firstly, taking derivative w.r.t x on both sides.

$$F'(x + y) = F'(x) + F'(x)F(y)$$

Substituting $x = 0$ and assuming $F'(0) = a$ such that $a \in \mathbb{R}$.

$$\frac{dF(y)}{dy} = a(1 + F(y)) \rightarrow \frac{dF(y)}{1 + F(y)} = a dy$$

Integrating on both sides while ignoring the integration constant results in the following assuming $e^a = b$:

$$\ln(1 + F(y)) = ay \rightarrow F(y) = e^{ay} - 1 = b^y - 1$$

The above is our final solution.

Example 3.2: Let, F be a differentiable function defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $N \in \mathbb{N}$ the following functional equation remains satisfied:

$$F\left(\frac{x + y}{N}\right) = \frac{F(x) + F(y)}{N}$$

To find all such applicable F .

Let us substitute $y = 0$ in the definition, which leads to:

$$F\left(\frac{x}{N}\right) = \frac{F(x)}{N}$$

Now, setting $x = u + v$ such that $u, v \in \mathbb{R}$ and using the definition.

$$F\left(\frac{u + v}{N}\right) = \frac{F(u + v)}{N} = \frac{F(u) + F(v)}{N}$$

Which ultimately leaves us with linear Cauchy equation. Now, taking a derivative w.r.t u .

$$\frac{d}{du}F(u + v) = \frac{d}{du}(F(u) + F(v)) \rightarrow F'(u + v) = F'(u)$$

Setting $u = 0$ and $F'(v) = a$ such that $a \in \mathbb{R}$ in the above expression gives us the following:

$$F'(v) = a \rightarrow dF(v) = a dv$$

Integrating on both sides results in the following:

$$F(v) = av$$

Example 3.3: Let, F be a differentiable function defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$F(x) + F(y) = F(x + y) - 2xy$$

Find all such applicable F .

We will be taking a derivative w.r.t x on both sides of the given equation.

$$\frac{d}{dx}(F(x) + F(y)) = \frac{d}{dx}(F(x + y) - 2xy)$$

$$F'(x) = F'(x + y) - 2y$$

On substituting $x = 0$ and rearranging the terms in the above expression, we have the following:

$$F'(y) - 2y = F'(0)$$

Do note that $F'(0) = 0$, because of which we have:

$$F'(y) = 2y$$

Integrating on both sides w.r.t y and ignoring the integration constant gives us our final answer.

$$F(y) = y^2$$

Example 3.4: Let, h be a differentiable function defined from $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$h(x) + h(y) = h(x + y) - 3xy(x + y)$$

Find all such applicable h .

As per usual, take a derivative on both sides w.r.t x .

$$\frac{d}{dx}(h(x) + h(y)) = \frac{d}{dx}(h(x + y) - 3xy(x + y))$$

$$h'(x) = h'(x + y) - 3y(x + y) - 3xy$$

$$h'(x) = h'(x + y) - 3xy - 3y^2$$

Substituting $x = 0$ in the above equation leads to the following;

$$h'(0) = h'(y) - 3y^2$$

Just like in the previous example, $h'(0) = 0$ in this case scenario as well.

$$h'(y) = 3y^2$$

Integrating on both sides w.r.t y and ignoring the integration constant gives us our final result;

$$h(y) = y^3$$

Example 3.5: Let, F be a differentiable function defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $A \in \mathbb{N}$ the following functional equation remains satisfied:

$$F(x + y) = A^{xy} F(x) F(y)$$

To find all such applicable F .

As per usual, take derivative on both sides w.r.t x .

$$\frac{dF(x + y)}{dx} = \frac{d}{dx} (A^{xy} F(x) F(y))$$

$$F'(x + y) = y \log A (A^{xy} F(x) F(y)) + A^{xy} F(x) F'(y)$$

On substituting $x = 0$ we obtain the following (Here, $F'(0) = 0$ and $F(0) = 1$):

$$F'(y) = y F(y) F(0) \log A + F(x) F'(0)$$

$$F'(y) = y F(y) \log A$$

$$\frac{dF(y)}{F(y)} = y \log A dy$$

Integrating on both sides and ignoring the integration constant gives us the following final result:

$$\log(F(y)) = \frac{y^2}{2} \log A = \log \left(A^{\frac{y^2}{2}} \right)$$

$$F(y) = A^{\frac{y^2}{2}}$$

Now, that we have understood how a differentiable solution looks like, we can proceed further and extrapolate on this concept to introduce twice differentiable functional solutions.

Let, F be a twice differentiable function defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied:

$$F(x + y) = F(x)F(y)$$

To find all such applicable F .

Firstly, we take a derivative on both sides w.r.t x .

$$F'(x + y) = F'(x)F(y)$$

As the function F is twice differentiable, we will take another derivative on both sides w.r.t x .

$$F''(x + y) = F''(x)F(y)$$

Substituting $x = 0$ in the above expression yields the following:

$$F''(y) = F''(0)F(y)$$

Let us assume $F''(0) = A$ such that $A \in \mathbb{R}$ because of which we have:

$$F''(y) - AF(y) = 0$$

Using operator theory, we can represent $F''(y) = D^2F(y)$ which allows us to factorize the above expression.

$$D^2F(y) - AF(y) = (D^2 - A)F(y) = (D - \sqrt{A})(D + \sqrt{A})F(y) = 0$$

Solving for each factor separately gives us the following results. As per usual, while integrating the integration constant is ignored and also here it is assumed that $e^{\sqrt{A}} = b$.

$$(D - \sqrt{A})F(y) = 0 \rightarrow \frac{dF(y)}{F(y)} = \sqrt{A} dy$$

$$\ln(F(y)) = y\sqrt{A} \rightarrow F(y) = \left(e^{\sqrt{A}}\right)^y = b^y$$

$$(D + \sqrt{A})F(y) = 0 \rightarrow \frac{dF(y)}{F(y)} = -\sqrt{A} dy$$

$$\ln(F(y)) = -y\sqrt{A} \rightarrow F(y) = \left(e^{\sqrt{A}}\right)^{-y} = b^{-y}$$

The final constructed solution has to amass the properties of both the solutions, but in this case the result becomes quite straight-forward to state:

$$F(y) = a^y$$

Such that, $a \in \mathbb{R}$.

Similarly, let us solve multiplicative Cauchy equation with the twice differentiable constraint as well.

Let, F be a twice differentiable function defined from $F : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied:

$$F(xy) = F(x)F(y)$$

Find all such applicable F .

As per usual, taking a derivative w.r.t x on both sides:

$$yF'(xy) = F'(x)F(y)$$

As F is twice differentiable, we take another derivative w.r.t x .

$$y^2F''(xy) = F''(x)F(y)$$

Substituting $x = 1$ in the above expression and also assuming that $F''(1) = b$ such that $b \in \mathbb{R}^+$.

$$y^2F''(y) = F''(1)F(y) = bF(y)$$

Using operator theory to write $F''(y) = D^2F(y)$ and factorize the resulting expression.

$$D^2F(y) - \frac{b}{y^2}F(y) = \left(D^2 - \frac{b}{y^2}\right)F(y) = \left(D - \frac{\sqrt{b}}{y}\right)\left(D + \frac{\sqrt{b}}{y}\right)F(y) = 0$$

Solving for each factor and during integration the integration constant is ignored as per usual.

$$\left(D - \frac{\sqrt{b}}{y}\right)F(y) = 0 \Rightarrow \frac{dF(y)}{F(y)} = \sqrt{b}\frac{dy}{y}$$

$$\log F(y) = \log(y^{\sqrt{b}}) \Rightarrow F(y) = y^{\sqrt{b}}$$

$$\left(D + \frac{\sqrt{b}}{y}\right)F(y) = 0 \Rightarrow \frac{dF(y)}{F(y)} = -\sqrt{b}\frac{dy}{y}$$

$$\log F(y) = \log(y^{-\sqrt{b}}) \Rightarrow F(y) = y^{-\sqrt{b}}$$

Notice that the power term in first and second solution together cover the entire real line and because of this fact, we state our general solution as follows:

$$F(y) = y^c$$

Such that $c \in \mathbb{R}$.

Let us solve an example before going to the next section.

Example 3.6: Let f be a twice differentiable function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied.

$$f(x + y) - 3xy(x + y) = f(x) + f(y)$$

Find all such applicable f .

Firstly, taking derivative on both sides w.r.t x .

$$f'(x + y) - 3y(x + y) - 3xy = f'(x)$$

Taking another derivate on both sides w.r.t x as f is a twice differentiable function.

$$f''(x + y) - 6y = f''(x)$$

Substituting $x = 0$ in the above expression to simplify further. (Note: Here it is assumed that $f''(0) = 0$, which can be proved quite easily by substituting $y = -x$ in first derivative expression and taking another derivative w.r.t x and then substituting $x = 0$)

$$f''(y) = 6y$$

Integrating the above expression twice and do note that $f(0) = f'(0) = 0$ can also be proven quite easily by referring and making use of the above derived expressions. Because of which, we have:

$$f'(y) = 3y^2 + f'(0) = 3y^2$$

$$f(y) = y^3 + f(0) = y^3$$

Therefore, our final solution is given as:

$$f(y) = y^3$$

Let us now move on to the next section where we will explore constraints that are more rigid than differentiability and their corresponding solutions to Cauchy equations.

Analytic Cauchy Equations

For a function to be analytic, it must have a corresponding Taylor series expansion. Let us understand more about this with the following example.

Let, f be an analytical function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied.

$$f(x + y) = f(x) + f(y)$$

To find all such applicable f .

As it is given to us that f is an analytical function, we can assume it to be of the following form:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)x^n}{n!} = \sum_{n=0}^{\infty} a_n x^n$$

Essentially, we are considering the Taylor series expansion of f around the neighbourhood of $x = 0$. It becomes safe to conclude that $a_n \in \mathbb{R}$. Using this series expression, we can state the following:

$$f(x + y) = \sum_{n=0}^{\infty} a_n (x + y)^n$$

$$f(x) + f(y) = \sum_{n=0}^{\infty} a_n (x^n + y^n)$$

Now, we have to figure out the numeric values of all a_n . This can be done with ease by comparing corresponding terms to a given a coefficient.

$$a_0 = 2a_0 \Rightarrow a_0 = 0$$

$$a_1(x) + a_1(y) = a_1(x + y) \Rightarrow a_1 \in \mathbb{R}$$

$$a_2(x^2) + a_2(y^2) = a_2(x^2 + y^2 + 2xy) \Rightarrow a_2 = 0$$

$$a_3(x^3) + a_3(y^3) = a_3(x^3 + y^3 + 3x^2y + 3xy^2) \Rightarrow a_3 = 0$$

Trivial solutions are obtained for all $n \geq 4$ as well:

$$a_n(x^n + y^n) = a_n \left(x^n + y^n + \sum_{k=1}^{n-1} \binom{n}{k} x^k y^{n-k} \right) \Rightarrow a_n = 0$$

Therefore, our final solution can be stated as such by substituting $a_i = 0$ for every $i \geq 0$ except $i = 1$.

$$f(x) = 0 + a_1x + (0)x^2 + (0)x^3 + \dots = a_1x$$

The above is the analytical solution of linear Cauchy equation. The case is quite similar with the simplified linear Pexider equation as well.

Let, F, G, H be analytical functions defined from $F, G, H : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied.

$$F(x + y) = G(x) + H(y)$$

To find all such applicable F, G, H .

Let the Taylor series expansions for F, G, H be written as such:

$$F(x) = \sum_{n=0}^{\infty} a_n x^n, G(x) = \sum_{n=0}^{\infty} b_n x^n, H(x) = \sum_{n=0}^{\infty} c_n x^n$$

Such that $a_n, b_n, c_n \in \mathbb{R}$, for every $n \geq 0$. Using the above series, we can rewrite the functional equation as follows:

$$F(x + y) = \sum_{n=0}^{\infty} a_n (x + y)^n$$

$$G(x) + H(y) = \sum_{n=0}^{\infty} b_n x^n + c_n y^n$$

On comparing coefficients for each term, we obtain the following results:

$$a_0 = b_0 + c_0$$

$$a_1(x + y) = b_1(x) + c_1(y) \Rightarrow a_1 = b_1, a_1 = c_1$$

$$a_2(x^2 + y^2 + 2xy) = b_2x^2 + c_2y^2 \Rightarrow a_2 = b_2 = c_2 = 0$$

$$a_3(x^3 + y^3 + 3x^2y + 3xy^2) = b_3x^3 + c_3y^3 \Rightarrow a_3 = b_3 = c_3 = 0$$

Similarly, for all $n \geq 4$ the following can be concluded:

$$a_n \left(x^n + y^n + \sum_{k=1}^{n-1} \binom{n}{k} x^k y^{n-k} \right) = b_n x^n + c_n y^n \Rightarrow a_n = b_n = c_n = 0$$

Which ultimately implies that the functions F, G, H are of the following form :

$$F(x) = a_0 + a_1 x + (0)x^2 + (0)x^3 + \dots = b_0 + c_0 + a_1 x$$

$$G(x) = b_0 + a_1 x + (0)x^2 + (0)x^3 + \dots = b_0 + a_1 x$$

$$H(x) = c_0 + a_1 x + (0)x^2 + (0)x^3 + \dots = c_0 + a_1 x$$

With that done, let us now work with the generalized linear Pexider equation.

Let, f, f_i be analytical functions defined from $f, f_i: \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x_i \in \mathbb{R}$ assuming $1 \leq i \leq n$, the following functional equation remains satisfied.

$$f \left(\sum_{i=1}^n x_i \right) = \sum_{i=1}^n f_i(x_i)$$

To find all such applicable f, f_i .

Let us now construct a generalized Taylor series expansion for f_i and f assuming $x \in \mathbb{R}$:

$$f_i(x) = \sum_{j=0}^{\infty} a_j^{(i)} x^j, f(x) = \sum_{j=0}^{\infty} a_j x^j$$

Thus, the given functional equation can be expressed as such using the above developed series :

$$f \left(\sum_{i=1}^n x_i \right) = \sum_{j=0}^{\infty} a_j \left(\sum_{i=1}^n x_i \right)^j$$

$$\sum_{i=1}^n f_i(x_i) = \sum_{i=1}^n \sum_{j=0}^{\infty} a_j^{(i)} x_i^j$$

Comparing the two expressions based on their coefficients results in the following:

$$a_0 = \sum_{i=1}^n a_0^{(i)}$$

$$a_1 \left(\sum_{i=1}^n x_i \right) = \sum_{i=1}^n a_1^{(i)} x_i \Rightarrow a_1 = a_1^{(i)}$$

$$a_2 \left(\sum_{i=1}^n x_i \right)^2 = \sum_{i=1}^n a_2^{(i)} x_i^2 \Rightarrow a_2 = a_2^{(i)} = 0$$

$$a_3 \left(\sum_{i=1}^n x_i \right)^3 = \sum_{i=1}^n a_3^{(i)} x_i^3 \Rightarrow a_3 = a_3^{(i)} = 0$$

Only trivial solutions are obtained for all $j \geq 4$. Let, $j = m$ such that $m \geq 4$.

$$a_m \left(\sum_{i=1}^n x_i \right)^m = \sum_{i=1}^n a_m^{(i)} x_i^m \Rightarrow a_m = a_m^{(i)} = 0$$

Ultimately, using the obtained coefficient values, we can confidently state the general solution for both f, f_i as such assuming $x \in \mathbb{R}$:

$$f(x) = a_0 + a_1 x + (0)x^2 + (0)x^3 + \dots = \sum_{i=1}^n a_0^{(i)} + a_1 x$$

$$f_i(x) = a_0^{(i)} + a_1^{(i)} x + (0)x^2 + (0)x^3 + \dots = a_0^{(i)} + a_1 x$$

Let us solve a few examples based on analyticity before moving to the next segment.

Example 4.1: Let h be an analytical function defined from $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $k \in \mathbb{N}$ the following functional equation remains satisfied.

$$h\left(\frac{x+y}{k}\right) = \frac{h(x) + h(y)}{k}$$

To find all such applicable h .

Let us assume the Taylor series expansion of h to be as follows:

$$h(x) = \sum_{n=0}^{\infty} b_n x^n$$

Such that $b_n \in \mathbb{R}$. By using the above established series, we can rewrite the functional equation as follows:

$$h\left(\frac{x+y}{k}\right) = \sum_{n=0}^{\infty} b_n \left(\frac{x+y}{k}\right)^n = \sum_{n=0}^{\infty} \frac{b_n}{k^n} (x+y)^n$$

$$\frac{h(x) + h(y)}{k} = \sum_{n=0}^{\infty} \frac{b_n}{k} (x^n + y^n)$$

On comparing the coefficients, we obtain the following:

$$b_0 = \frac{2b_0}{k} \rightarrow b_0 = 0$$

$$\frac{b_1}{k} (x+y) = \frac{b_1}{k} (x) + \frac{b_1}{k} (y) \rightarrow b_1 \in \mathbb{R}$$

$$\frac{b_2}{k^2} (x+y)^2 = \frac{b_2}{k} (x^2 + y^2) \rightarrow b_2 = 0$$

Similarly, for $n \geq 3$ values of all obtained coefficients are trivial. Let, $n = m$ such that $m \geq 3$.

$$\frac{b_m}{k^m} (x+y)^m = \frac{b_m}{k} (x^m + y^m) \rightarrow b_m = 0$$

Therefore, our final solution is given as follows.

$$h(x) = 0 + b_1(x) + (0)x^2 + \dots = b_1(x)$$

Example 4.2: Let F be an analytical function defined from $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied.

$$F(x+y) = e^y F(x) + e^x F(y)$$

To find all such applicable F .

Firstly, dividing the term e^{x+y} on both sides.

$$e^{-x-y} F(x+y) = e^{-x} F(x) + e^{-y} F(y)$$

Now it becomes important realize that as $F(x)$ is an analytical function and e^{-x} is an analytical function, the product of two analytical functions $e^{-x}F(x)$ will also be an analytical function. We will rename the product function $H(x) = e^{-x}F(x)$, by which we have:

$$H(x + y) = H(x) + H(y)$$

Here, we assume the Taylor series of H to be as follows:

$$H(x) = \sum_{n=0}^{\infty} B_n x^n$$

Because of which the derived functional equation transforms into the following:

$$\begin{aligned} H(x + y) &= \sum_{n=0}^{\infty} B_n (x + y)^n \\ H(x) + H(y) &= \sum_{n=0}^{\infty} B_n (x^n + y^n) \end{aligned}$$

Comparing coefficients gives us the following results:

$$\begin{aligned} B_0 &= 2B_0 \Rightarrow B_0 = 0 \\ B_1(x + y) &= B_1(x) + B_1(y) \Rightarrow B_1 \in \mathbb{R} \\ B_2(x + y)^2 &= B_2x^2 + B_2y^2 \Rightarrow B_2 = 0 \end{aligned}$$

And similarly, let $m = n$ such that $n \geq 3$ and we obtain only trivial solutions.

$$B_n(x + y)^n = B_nx^n + B_ny^n \Rightarrow B_n = 0$$

By substituting values of all obtained coefficients, we can construct the general solution of H .

$$H(x) = 0 + B_1x + (0)x^2 + \dots = B_1x$$

Therefore, the final solution of F is given as follows:

$$F(x) = H(x)e^x = B_1xe^x$$

That concludes the Analytical Cauchy equations section and in the next section we will start to assume that the function is transformable which is an even more robust condition than analyticity.

Transformable Cauchy Equations

To start things off, it would prove to be quite helpful to understand integral transforms and namely, Laplace transform. All key results and properties are discussed in the appendix section.

Firstly, Laplace transform in one dimension is stated as follows:

$$L_x\{f(x)\}(p) = \int_0^{\infty} e^{-px} f(x) dx = F(p)$$

Similarly, Laplace transform in two dimensions is described as such:

$$L_{x,y}\{f(x,y)\}(p,q) = \int_0^{\infty} \int_0^{\infty} e^{-px-xy} f(x,y) dx dy = \bar{F}(p,q)$$

This transform is quite helpful in solving Cauchy equations and we will understand the reason behind it soon, but before that we need to derive some important results which will aid us greatly in solving functional equations in this section.

Firstly calculating $L_{x,y}\{f(x)\}(p,q)$ and $L_{x,y}\{f(y)\}(p,q)$:

$$L_{x,y}\{f(x)\}(p,q) = \int_0^{\infty} \int_0^{\infty} e^{-px-xy} f(x) dx dy = \int_0^{\infty} e^{-xy} \int_0^{\infty} e^{-px} f(x) dx dy$$

$$L_{x,y}\{f(x)\}(p,q) = \int_0^{\infty} e^{-xy} F(p) dy = \frac{F(p)}{q}$$

$$L_{x,y}\{f(y)\}(p,q) = \int_0^{\infty} \int_0^{\infty} e^{-px-xy} f(y) dx dy = \frac{1}{p} \int_0^{\infty} e^{-xy} f(y) dy$$

$$L_{x,y}\{f(y)\}(p,q) = \frac{F(q)}{p}$$

Now, moving forward and calculating $L_{x,y}\{f(x+y)\}(p,q)$:

$$L_{x,y}\{f(x+y)\}(p,q) = \int_0^\infty \int_0^\infty e^{-px-xy} f(x+y) dx dy$$

Assume $u = x + y$ and $du = dx$, because of which we have:

$$L_{x,y}\{f(x+y)\}(p,q) = \int_0^\infty \int_y^\infty e^{-p(u)-(q-p)y} f(u) du dy$$

Now, we change order of integration by using the Fubini's theorem. Notice that, $0 \leq y < \infty$ and $y \leq u < \infty$ and by flipping the order of integration, we observe that $0 \leq u < \infty$ and $0 \leq y < u$, because of which we have the following:

$$L_{x,y}\{f(x+y)\}(p,q) = \int_0^\infty e^{-pu} f(u) \int_0^u e^{-(q-p)y} dy du$$

$$L_{x,y}\{f(x+y)\}(p,q) = \int_0^\infty e^{-pu} f(u) \left[\frac{1 - e^{-(q-p)u}}{q-p} \right] du$$

$$L_{x,y}\{f(x+y)\}(p,q) = \frac{1}{q-p} \int_0^\infty e^{-pu} f(u) du - \frac{1}{q-p} \int_0^\infty e^{-qu} f(u) du$$

Therefore, the final solution is given as:

$$L_{x,y}\{f(x+y)\}(p,q) = \frac{F(p) - F(q)}{q-p}$$

We now have the required results in our inventory to solve Cauchy equations using Laplace transforms. Let us first solve Linear Cauchy equation using two-dimensional Laplace transform.

Let, f be a well-defined Laplace transformable function defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(x+y) = f(x) + f(y)$$

Find all such applicable f .

We will be taking two-dimensional Laplace transform on both sides and referring to the previously derived results, obtaining the following:

$$L_{x,y}\{f(x+y)\}(p,q) = L_{x,y}\{f(x)\}(p,q) + L_{x,y}\{f(y)\}(p,q)$$

$$\frac{F(p) - F(q)}{q - p} = \frac{F(p)}{q} - \frac{F(q)}{p}$$

Simplifying the above expression further:

$$pq(F(p) - F(q)) = (q - p)(pF(p) - qF(q))$$

$$pqF(p) - pqF(q) = qpF(p) - q^2F(q) - p^2F(p) - pqF(q)$$

$$p^2F(p) = q^2F(q)$$

Notice that the right-hand side expression does not depend on p and because of which we can say that $q^2F(q) = c$ such that $c \in \mathbb{R}$, by which we have:

$$p^2F(p) = c \Rightarrow F(p) = \frac{c}{p^2}$$

We take L_x^{-1} on both sides and we obtain the following (Refer to the appendix to find all key Laplace transform results):

$$L_x^{-1}\{F(P)\} = L_x^{-1}\left\{\frac{c}{p^2}\right\} \Rightarrow f(x) = cx$$

Next up, we will be solving exponential Cauchy equation using two-dimensional Laplace transform.

Let, f be a well-defined function which has a finite two-dimensional Laplace transform defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(x+y) = f(x)f(y)$$

To find all such applicable f .

Similar to our previous exposition, taking Laplace transform on both sides.

$$L_{x,y}\{f(x+y)\}(p,q) = L_{x,y}\{f(x)f(y)\}(p,q)$$

$$\frac{F(p) - F(q)}{q - p} = \frac{F(p)}{q} \frac{F(q)}{p}$$

Simplifying the above expression further:

$$\begin{aligned} pq(F(p) - F(q)) &= (q - p)pqF(p)F(q) \\ pqF(p) - pqF(q) &= p^2qF(p) - pq^2F(q) \\ F(p)(pq - p^2q) &= F(q)(pq - pq^2) \\ F(p)(1 - p) &= F(q)(1 - q) \end{aligned}$$

As right-hand side expression does not contain p , we can safely assume that $F(q)(1 - q) = -c$ such that $c \in \mathbb{R}$.

$$F(p)(1 - p) = -c \Rightarrow F(p) = \frac{-c}{1 - p} = \frac{c}{p - 1}$$

Taking L_x^{-1} on both sides gives us the final result:

$$L_x^{-1}\{F(p)\} = L_x^{-1}\left\{\frac{c}{p - 1}\right\} \Rightarrow f(x) = ce^x$$

Similarly, the exponential Cauchy equation can also be solved using double-sided Laplace transform. A double-sided Laplace transform is given as follows:

$$L_x^*\{f(x)\}(p) = \int_{\mathbb{R}} e^{-px} f(x) dx = \tilde{f}(p)$$

calculating $L_x^*\{f(x + y)\}(p)$ as it will aid us in solving the exponential Cauchy equation later.

$$L_x^*\{f(x + y)\}(p) = \int_{\mathbb{R}} e^{-px} f(x + y) dx$$

Let $u = x + y$, such that $du = dx$ by which we have:

$$L_x^*\{f(x + y)\}(p) = \int_{\mathbb{R}} e^{-p(u-y)} f(u) du = e^{py} \int_{\mathbb{R}} e^{-pu} f(u) du = e^{py} \tilde{f}(p)$$

Now, we can work over exponential Cauchy equation using double-sided Laplace transform.

Let, f be a well-defined function which has a finite and non-zero double-sided Laplace transform defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following functional equation remains satisfied.

$$f(x + y) = f(x)f(y)$$

To find all such applicable f .

Taking L_x^* on both sides resulting in the following:

$$L_x^*\{f(x + y)\}(p) = L_x^*\{f(x)f(y)\}(p)$$

$$e^{py}\tilde{f}(p) = f(x)\tilde{f}(p)$$

Dividing $\tilde{f}(p)$ on both sides as it is given that $\tilde{f}(p) \neq 0$.

$$f(y) = e^{py}$$

Assume that $p = a$ such that $a \in \mathbb{R}$ and $e^a = b$, by which we obtain our final result.

$$f(y) = e^{ay} = (e^a)^y = b^y$$

Now, we will shift our focus to multiplicative Cauchy equation which can be solved with ease using the Mellin transform. Mellin transform's definition is given as such:

$$M_x\{f(x)\}(s) = \int_0^\infty x^{s-1}f(x) dx = \psi(s)$$

Such that analytical strip of s is given as $\langle a_f, b_f \rangle$, where both $a_f, b_f \in \mathbb{R}$.

Using this definition, let us calculate the value of $M_x\{f(yx)\}(s)$:

$$M_x\{f(yx)\}(s) = \int_0^\infty x^{s-1}f(yx) dx$$

Let, $u = yx$ and $du = y dx$ by which we have:

$$M_x\{f(yx)\}(s) = \int_0^\infty \left(\frac{u}{y}\right)^{s-1} f(u) \frac{du}{y} = y^{-s} \int_0^\infty u^{s-1} f(u) du = y^{-s} \psi(s)$$

Using this information, we can now solve the multiplicative Cauchy equation using Mellin transform.

Let, f be a well-defined function which has a finite and non-zero Mellin transform defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(xy) = f(x)f(y)$$

To find all such applicable f .

Taking Mellin transform on both sides leads to the following expression:

$$M_x\{f(xy)\}(s) = M_x\{f(x)f(y)\}(s)$$

$$y^{-s}\psi(s) = \psi(s)f(y)$$

Cancelling the term $\psi(s)$ on both sides as it is given that $\psi(s) \neq 0$.

$$f(y) = y^{-s}$$

Now, we assume that $s = -c$ such that $c \in \mathbb{R}$, which gives us our final solution.

$$f(y) = y^c$$

With that done, let us solve a few examples based on the learned transforms.

Example 5.1: Let, f be a well-defined function which has a finite two-dimensional Laplace transform defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(x + y) = f(x) + f(y)$$

To find all such applicable f .

We have already solved this example previously but there is yet another simpler route to obtain the final solution. Let us firstly take $L_{x,y}$ on both sides.

$$L_{x,y}\{f(x + y)\}(p, q) = L_{x,y}\{f(x)\}(p, q) + L_{x,y}\{f(y)\}(p, q)$$

$$\frac{F(p) - F(q)}{q - p} = \frac{F(p)}{q} + \frac{F(q)}{p}$$

Here comes the crucial step, assume that $p \rightarrow q$ because of which we have the following simplified expression:

$$\lim_{p \rightarrow q} \left\{ \frac{F(p) - F(q)}{q - p} \right\} = \lim_{p \rightarrow q} \left\{ \frac{F(p)}{q} + \frac{F(q)}{p} \right\}$$

$$-F'(p) = 2 \frac{F(p)}{p}$$

Now, we solve the above ordinary differential equation assuming $c \in \mathbb{R}^+$ and $e^c = C$.

$$-\frac{dF(p)}{F(p)} = -2 \frac{dp}{p} \rightarrow -\ln F(p) = -2 \ln(p) + c$$

$$\ln F(p) = \ln(p^{-2}) + c$$

$$F(p) = e^{\ln(p^{-2})+c} = \frac{C}{p^2}$$

Taking L_x^{-1} on both sides gives us our final result:

$$L_x^{-1}\{F(p)\} = L_x^{-1}\left\{\frac{C}{p^2}\right\} \rightarrow f(x) = Cx$$

Such that $C, x \in \mathbb{R}^+$.

Example 5.2: Let, f be a well-defined function which has a finite and non-zero Mellin transform defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ and $k \in \mathbb{N}$ the following functional equation remains satisfied.

$$f(kxy) = kf(x)f(y)$$

To find all such applicable f .

Firstly, taking M_x on both sides resulting in the following:

$$M_x\{f(kxy)\}(s) = M_x\{kf(x)f(y)\}(s)$$

$$(ky)^{-s}\psi(s) = kf(y)\psi(s)$$

Cancelling $\psi(s)$ on both sides of the equation and assuming $s = -c$ such that $c \in \mathbb{R}$, which leads to the following:

$$kf(y) = (ky)^c \rightarrow f(y) = k^{c-1}y^c$$

The above is our final solution.

Example 5.3: Let, f be a well-defined function which has a finite and non-zero double-sided Laplace transform defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ and $k \in \mathbb{N}$ the following functional equation remains satisfied.

$$f(x + y) = f(x + k)f(y - k)$$

To find all such applicable f .

Taking L_x^* on both sides of the equation.

$$L_x^*\{f(x + y)\}(p) = L_x^*\{f(x + k)f(y - k)\}(p)$$

$$e^{py}\tilde{f}(p) = e^{ky}\tilde{f}(p)f(y - k)$$

Cancelling $\tilde{f}(p)$ on both sides as $\tilde{f}(p) \neq 0$ and then rearranging the terms.

$$f(y - k) = e^{p(y-k)}$$

Here, we can assume that $y - k = w$ such that $w \in \mathbb{R}$.

$$f(w) = e^{pw}$$

The above is our final solution.

Example 5.4: Let, f be a well-defined function which has a finite and non-zero two-dimensional Laplace transform defined from $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}^+$ the following functional equation remains satisfied.

$$f(x - y) = f(x) - f(y)$$

Assume that f is an odd function, find all such applicable f .

We will be taking $L_{x,y}$ on both sides and do note that $L_{x,y}\{f(x - y)\}(p, q)$ has the following result (Refer to the appendix).

$$L_{x,y}\{f(x - y)\}(p, q) = \frac{F(p) - F(q)}{p + q}$$

$$L_{x,y}\{f(x - y)\}(p, q) = L_{x,y}\{f(x)\}(p, q) - L_{x,y}\{f(y)\}(p, q)$$

$$\frac{F(p) - F(q)}{p + q} = \frac{F(p)}{q} - \frac{F(q)}{p}$$

Simplifying the above derived algebraic expression.

$$qp(F(p) - F(q)) = (p + q)(pF(p) - qF(q))$$

$$qpF(p) - qpF(q) = p^2F(p) - pqF(q) + pqF(q) - q^2F(q)$$

$$p^2F(p) - q^2F(q) = 0$$

$$p^2F(p) = q^2F(q)$$

As both sides of the above expression are unrelated to each other, we assume $q^2F(q) = c$ such that $c \in \mathbb{R}$, by which we have:

$$p^2F(p) = c \rightarrow F(p) = \frac{c}{p^2}$$

Taking L_x^{-1} on both sides gives us our final result.

$$L_x^{-1}\{F(p)\} = L_x^{-1}\left\{\frac{c}{p^2}\right\} \rightarrow f(x) = cx$$

With that we are done with the transformable Cauchy equations. Next up, we will be discussing two-dimensional Cauchy equations in a continuous frame.

Two-dimensional Cauchy equations

So far, we have discussed Cauchy equations defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ or $f : \mathbb{R}^+ \rightarrow \mathbb{R}$. But a natural question arises, what if the input dimensionality was $\mathbb{R}^2$ or $\mathbb{R}^3$ instead? To respond to that curiosity, we will be discussing two-dimensional Cauchy equations at length before working with its generalized dimensional variant in the next section. Let us proceed with the derivation of the linear Cauchy equation in two-dimensions.

Let, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x_1, x_2, y_1, y_2 \in \mathbb{R}$, the following functional equation remains satisfied:

$$f(x_1 + x_2, y_1 + y_2) = f(x_1, y_1) + f(x_2, y_2)$$

To find all such applicable f .

To start with this derivation, we can extrapolate the given equation assuming $\forall x_i, y_i \in \mathbb{R}$ such that $1 \leq i \leq n$ where n is some natural number greater than 2.

$$f\left(\sum_{1 \leq i \leq n} x_i, \sum_{1 \leq i \leq n} y_i\right) = \sum_{1 \leq i \leq n} f(x_i, y_i)$$

Here, we set $x_i = x$ and $y_i = y$ for every $1 \leq i \leq n$. Because of which, we have:

$$f(nx, ny) = nf(x, y)$$

Let us now say that $x = \frac{mt}{n}$ and $y = \frac{mp}{n}$ where $m \in \mathbb{N}$ and $t, p \in \mathbb{R}$.

$$f(mt, mp) = nf\left(\frac{mt}{n}, \frac{mp}{n}\right)$$

By using the definition itself we can rewrite the above expression as:

$$f\left(\frac{mt}{n}, \frac{mp}{n}\right) = \frac{m}{n} f(t, p)$$

Here, we make yet another assumption by saying $q = \frac{m}{n}$ such that $q \in \mathbb{Q}$.

$$f(qt, qp) = qf(t, p)$$

Considering a sequence of a rational numbers $\{q\}$, we note its limiting behaviour to be as follows:

$$\lim_{n \rightarrow \infty} \{q_n\} = r$$

Such that $r \in \mathbb{R}$. This allows us to rewrite our above derived expression.

$$\lim_{n \rightarrow \infty} f(q_n t, q_n p) = f(rt, rp) = rf(t, p)$$

Here, we substitute $t = 0, p = 1$ and then substitute $t = 1, p = 0$ by taking $r = r'$ to generalize our final result. We also assume $f(1, 0) = A$ and $f(0, 1) = B$

$$f(r, 0) = rf(1, 0) = Ar$$

$$f(0, r') = r'f(0, 1) = Br'$$

Adding up these two equations gives us our final result:

$$f(r, 0) + f(0, r') = f(r, r') = Ar + Br'$$

Such that $A, B, r, r' \in \mathbb{R}$.

By using this result, we can derive the final solution of simplified two-dimensional linear Pexider equation quite easily too.

Let, $F, G, H : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that F, G, H are continuous functions and $\forall x, y, u, v \in \mathbb{R}$, the following functional equation remains satisfied:

$$F(x + y, u + v) = G(x, u) + H(y, v)$$

To find all such applicable F, G, H .

Firstly, setting $x = 0, u = 0$ and then setting $y = 0, v = 0$ gives us the following system of equations:

$$F(x, u) = G(x, u) + H(0, 0)$$

$$F(y, v) = G(0, 0) + H(y, v)$$

Adding these two equations results in the following:

$$F(x, u) + F(y, v) = (G(x, u) + H(y, v)) + (G(0, 0) + H(0, 0))$$

$$F(x, u) + F(y, v) = F(x + y, u + v) + F(0, 0)$$

Assuming $I(x, u) = F(x, u) - F(0, 0)$ results in a linear two-dimensional Cauchy equation.

$$I(x + y, u + v) = I(x, u) + I(y, v)$$

Referring to our previous exposition, we can directly state the observed results as follows:

$$I(x, u) = Ax + Bu \Rightarrow F(x, u) = Ax + Bu + F(0, 0)$$

Such that $A, B \in \mathbb{R}$. Now, if we assume $G(0, 0) = a$ and $H(0, 0) = b$ and as a consequence $F(0, 0) = a + b$, we obtain the following final result:

$$F(x, u) = Ax + Bu + a + b$$

$$G(x, u) = Ax + Bu + a$$

$$H(x, u) = Ax + Bu + b$$

Goes without saying, $a, b \in \mathbb{R}$. Let us now explore the two-dimensional exponential Cauchy equation.

Let, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x_1, x_2, y_1, y_2 \in \mathbb{R}$, the following functional equation remains satisfied:

$$f(x_1 + x_2, y_1 + y_2) = f(x_1, y_1)f(x_2, y_2)$$

To find all such applicable f .

As per usual, we extrapolate the original function up assuming $\forall x_i, y_i \in \mathbb{R}$ such that $1 \leq i \leq n$ and $n \in \mathbb{N}_{\geq 2}$.

$$f\left(\sum_{1 \leq i \leq n} x_i, \sum_{1 \leq i \leq n} y_i\right) = \prod_{1 \leq i \leq n} f(x_i, y_i)$$

Here, we say that $x_i = x$ and $y_i = y$ for every $1 \leq i \leq n$ as $x, y \in \mathbb{R}$.

$$f(nx, ny) = f(x, y)^n$$

Now, we assume $x = \frac{mt}{n}, y = \frac{mp}{n}$ such that $m \in \mathbb{N}$ and $t, p \in \mathbb{R}$.

$$f(mt, mp) = f(t, p)^m = f\left(\frac{mt}{n}, \frac{mp}{n}\right)^n$$

By dealing with the powers in the above expression, we arrive at this crucial statement.

$$f\left(\frac{mt}{n}, \frac{mp}{n}\right) = f(t, p)^{m/n}$$

Assuming $\frac{m}{n} = q$ such that $q \in \mathbb{Q}$.

$$f(qt, qp) = f(t, p)^q$$

Considering a sequence of rational numbers $\{q\}$, we note the limiting behaviour of this sequence as follows:

$$\lim_{n \rightarrow \infty} \{q_n\} = r$$

Such that $r \in \mathbb{R}$. By using this, we rewrite our derived expression.

$$f(q_n t, q_n p) = f(t, p)^{q_n}$$

$$\lim_{n \rightarrow \infty} \{f(q_n t, q_n p)\} = f(rt, rp) = f(t, p)^r$$

Now, we establish a system of equations by firstly substituting $t = 1, p = 0$ and then substituting $t = 0, p = 1$ by taking $r = r'$ which helps in generalizing our final result.

$$f(r, 0) = f(1, 0)^r$$

$$f(0, r') = f(0, 1)^{r'}$$

Assuming $f(1, 0) = A$ and $f(0, 1) = B$ such that $A, B \in \mathbb{R}$. We multiply the resulting equations to obtain our final result:

$$f(r, r') = A^r B^{r'}$$

By using this very result, we can also find the solution to simplified exponential Pexider equation.

Let, $F, G, H : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that F, G, H are continuous functions and $\forall x, y, u, v \in \mathbb{R}$, the following functional equation remains satisfied:

$$F(x + u, y + v) = G(x, y)H(u, v)$$

To find all such applicable F, G, H .

Substituting $u = 0, v = 0$ and then substituting $x = 0, y = 0$ to obtain the following system of equations:

$$F(x, y) = G(x, y)H(0, 0)$$

$$F(u, v) = G(0, 0)H(u, v)$$

Multiplying the resulting equations together to obtain the following:

$$F(x, y)F(u, v) = (G(x, y)H(u, v))(G(0, 0)H(0, 0))$$

$$F(x, y)F(u, v) = F(x + u, y + v)F(0, 0)$$

Assuming $F(0, 0) \neq 0$ we divide this term on both sides of the expression and also assume that $f(x, y) = \frac{F(x, y)}{F(0, 0)}$, by which we have:

$$f(x, y)f(u, v) = f(x + u, y + v)$$

Using our obtained solution from the previous exposition yields the following assuming $F(0, 0) = \frac{1}{c}$ such that $c \in \mathbb{R} / \{0\}$:

$$f(x, y) = A^x B^y \rightarrow F(x, y) = cA^x B^y$$

Ultimately, the final solution can be stated as such:

$$F(x, y) = abA^x B^y, G(x, y) = aA^x B^y, H(x, y) = bA^x B^y$$

Here, we have assumed that $H(0, 0) = b$ and $G(0, 0) = a$ such that $a, b \in \mathbb{R}$ which itself implies that $F(0, 0) = c = ab$.

Moving forward, we will be discussing two-dimensional logarithmic Cauchy equation.

Let, $f : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x_1, x_2, y_1, y_2 \in \mathbb{R}^+$, the following functional equation remains satisfied:

$$f(x_1x_2, y_1y_2) = f(x_1, y_1) + f(x_2, y_2)$$

To find all such applicable f .

As per usual, extrapolating the functional equation to obtain the following:

$$f\left(\prod_{1 \leq i \leq n} x_i, \prod_{1 \leq i \leq n} y_i\right) = \sum_{1 \leq i \leq n} f(x_i, y_i)$$

Such that $\forall x_i, y_i \in \mathbb{R}^+$ given $1 \leq i \leq n$ (Where n is a positive integer greater than or equal to 2). Now, we set $x_i = x, y_i = y$ assuming $x, y \in \mathbb{R}^+$.

$$f(x^n, y^n) = nf(x, y)$$

Here, we say that $x = t^{\frac{m}{n}}$ and $y = p^{\frac{m}{n}}$ such that $m \in \mathbb{N}$ and $t, p \in \mathbb{R}^+$.

$$f\left(\left(t^{\frac{m}{n}}\right)^n, \left(p^{\frac{m}{n}}\right)^n\right) = f(t^m, p^m) = nf\left(t^{\frac{m}{n}}, p^{\frac{m}{n}}\right)$$

By making use of the definition, we can simplify the above expression as follows:

$$f\left(t^{\frac{m}{n}}, p^{\frac{m}{n}}\right) = \frac{m}{n} f(t, p)$$

By renaming $q = \frac{m}{n}$ such that $q \in \mathbb{Q}$, we can restate the above expression.

$$f(t^q, p^q) = qf(t, p)$$

Considering a sequence of rational numbers $\{q\}$, we note its limiting behaviour to be as follows:

$$\lim_{n \rightarrow \infty} \{q_n\} = r$$

Such that $r \in \mathbb{R}^+$. By making use of this limiting behaviour, we can generalize the above derived expression;

$$f(t^{q_n}, p^{q_n}) = q_n f(t, p)$$

$$\lim_{n \rightarrow \infty} \{f(t^{q_n}, p^{q_n})\} = f(t^r, p^r) = r f(t, p)$$

Firstly, we assume $p = P$ and $t = T$ such that $P, T \in \mathbb{R}^+$. Secondly, substituting $T = 1, P^r = A$ and then substituting $T^{r'} = B, P = 1$ while setting $r = r'$ helps in forming the given system of equations.

$$f(1, A) = \log_P A f(1, P)$$

$$f(B, 1) = \log_P B f(T, 1)$$

Adding these equations together results in the following:

$$f(B, A) = f(1, P) \log_P A + f(T, 1) \log_P B$$

Ultimately, the final solution is stated as such:

$$f(B, A) = b \log_P B + a \log_P A = \log_{P^{\frac{1}{b}}} B + \log_{P^{\frac{1}{a}}} A$$

$$f(B, A) = \log_D B + \log_C A$$

Here, it is assumed that $f(1, P) = a, f(T, 1) = b$ (needless to say, $a, b \in \mathbb{R}^+$) and it's also assumed that $P^{\frac{1}{a}} = C$ and $P^{\frac{1}{b}} = D$ such that $A, B, C, D \in \mathbb{R}^+$.

Using this solution, we can also quickly find the general solution of simplified logarithmic Pexider equation.

Let, $F, G, H : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ such that F, G, H are continuous functions and $\forall x, y, u, v \in \mathbb{R}^+$, the following functional equation remains satisfied:

$$F(xu, yv) = G(x, y) + H(u, v)$$

To find all such applicable F, G, H .

Substituting $u = 1, v = 1$ and then substituting $x = 1, y = 1$ results in the following system of equations:

$$F(x, y) = G(x, y) + H(1, 1)$$

$$F(u, v) = G(1, 1) + H(u, v)$$

Adding these two equations results in the following:

$$F(x, y) + F(u, v) = (G(x, y) + H(u, v)) + (G(1, 1) + H(1, 1))$$

$$F(x, y) + F(u, v) = F(xu, yv) + F(1, 1)$$

Now, we set $f(x, y) = F(x, y) - F(1, 1)$, by which we obtain two-dimensional logarithmic Cauchy equation.

$$f(x, y) + f(u, v) = f(xu, yv)$$

The solution to the above equation is given as follows:

$$f(x, y) = \log_C x + \log_D y \rightarrow F(x, y) = \log_C x + \log_D y + B$$

Such that $C, D, B \in \mathbb{R}^+$ and $F(1,1) = B$. Ultimately, the general solution of this equation is stated as such:

$$F(x, y) = \log_C x + \log_D y + a + b$$

$$G(x, y) = \log_C x + \log_D y + a$$

$$H(x, y) = \log_C x + \log_D y + b$$

Here, $G(1,1) = a$ and $H(1,1) = b$ which basically implies $B = a + b$ such that $a, b \in \mathbb{R}^+$.

Moving on, we now discuss two-dimensional multiplicative Cauchy equation.

Let, $f : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ such that f is a continuous function and $\forall x_1, x_2, y_1, y_2 \in \mathbb{R}^+$, the following functional equation remains satisfied:

$$f(x_1 x_2, y_1 y_2) = f(x_1, y_1) f(x_2, y_2)$$

To find all such applicable f .

Extrapolating the given functional equation by assuming $\forall x_i, y_i \in \mathbb{R}^+$ such that $1 \leq i \leq n$ (Where n is a natural number greater than or equal to 2).

$$f\left(\prod_{1 \leq i \leq n} x_i, \prod_{1 \leq i \leq n} y_i\right) = \prod_{1 \leq i \leq n} f(x_i, y_i)$$

Here, we set $x_i = x, y_i = y$ where $x, y \in \mathbb{R}^+$.

$$f(x^n, y^n) = f(x, y)^n$$

Now, we say that $x = t^{\frac{m}{n}}$ and $y = p^{\frac{m}{n}}$ such that $m \in \mathbb{N}$ and $t, p \in \mathbb{R}^+$.

$$f\left(\left(t^{\frac{m}{n}}\right)^n, \left(p^{\frac{m}{n}}\right)^n\right) = f(t, p)^m = f\left(t^{\frac{m}{n}}, p^{\frac{m}{n}}\right)^n$$

On scaling the powers appropriately, we obtain the following:

$$f\left(t^{\frac{m}{n}}, p^{\frac{m}{n}}\right) = f(t, p)^{\frac{m}{n}}$$

It becomes safe to assume that $q = \frac{m}{n}$ such that $q \in \mathbb{Q}$, because of which we have:

$$f(t^q, p^q) = f(t, p)^q$$

Assuming a sequence of rational numbers $\{q\}$ where we note its limiting behaviour to be as follows;

$$\lim_{n \rightarrow \infty} \{q_n\} = r$$

Such that $r \in \mathbb{R}^+$. Using this limiting behaviour, we can simplify our above derived expression.

$$f(t^{q_n}, p^{q_n}) = f(t, p)^{q_n}$$

$$\lim_{n \rightarrow \infty} \{f(t^{q_n}, p^{q_n})\} = f(t^r, p^r) = f(t, p)^r$$

Now, we set $t^r = T$ and $p^{r'} = P$ such that $T, P \in \mathbb{R}^+$. Substituting $P = 1$ and then setting $r = r'$ and substituting $T = 1$ assuming $f(T, 1) = A$ and $f(1, P) = B$ gives us the following system of equations.

$$f(T, 1) = f(t, 1)^r = A^{\log_t T} = (A^{\log_A T})^{\frac{1}{\log_A t}} = T^{\log_t A} = T^a$$

$$f(1, P) = f(1, p) = B^{r'} = B^{\log_p P} = (B^{\log_B P})^{\frac{1}{\log_B p}} = P^{\log_B p} = P^b$$

Such that $a = \log_t A, b = \log_B p$ such that $a, b \in \mathbb{R}$. Therefore, final solution can be stated as such:

$$f(T, 1)f(1, P) = f(T, P) = T^a P^b$$

Let us now find the general solution of the simplified two-dimensional multiplicative Pexider equation.

Let, $F, G, H : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ such that F, G, H are continuous functions and $\forall x, y, u, v \in \mathbb{R}^+$, the following functional equation remains satisfied:

$$F(xu, yv) = G(x, y)H(u, v)$$

To find all such applicable F, G, H .

Substituting $u, v = 1$ and then substituting $x, y = 1$ to obtain the following system of equations:

$$F(x, y) = G(x, y)H(1, 1)$$

$$F(u, v) = G(1, 1)H(u, v)$$

Multiplying these equations together results in the following:

$$F(x, y)F(u, v) = (G(x, y)H(u, v))(G(1, 1)H(1, 1))$$

$$F(x, y)F(u, v) = F(xu, yv)F(1, 1)$$

Needless to say, that $F(1, 1) \neq 0$. Also, we can assume that $\frac{F(x, y)}{F(1, 1)} = f(x, y)$ because of which we have:

$$f(x, y)f(u, v) = f(xu, yv)$$

The general solution of the above expression is given as follows assuming $F(1, 1) = \frac{1}{c}$ such that $c \in \mathbb{R} / \{0\}$.

$$f(x, y) = x^a y^b \rightarrow F(x, y) = cx^a y^b$$

Therefore, general solution of all of the required functions is stated as such:

$$F(x, y) = ABx^a y^b, G(x, y) = Ax^a y^b, H(x, y) = Bx^a y^b$$

Such that $G(1, 1) = A, H(1, 1) = B$ (needless to say, $A, B \in \mathbb{R}$) which ultimately implies $F(1, 1) = c = AB$. With that, we have discussed two-dimensional Cauchy equations in detail and we are all equipped to study the general n -dimensional variant but before that, let us solve few examples.

Example 6.1: Let, $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that F is a continuous function and $\forall x, y, u, v \in \mathbb{R} \ \& \ N \in \mathbb{N}$ the following functional equation remains satisfied:

$$F\left(\frac{x+u}{N}, \frac{y+v}{N}\right) = \frac{F(x, y) + F(u, v)}{N}$$

To find all such applicable functions F .

Let us substitute $u = v = 0$ by which we have the following:

$$F\left(\frac{x}{N}, \frac{y}{N}\right) = \frac{F(x, y) + F(0, 0)}{N}$$

Now, we assume that $x = t + p$ and $y = q + s$ such that $t, p, q, s \in \mathbb{R}$. On using this information, we have:

$$F\left(\frac{t+p}{N}, \frac{q+s}{N}\right) = \frac{F(t, q) + F(p, s)}{N} = \frac{F(t+p, q+s) + F(0, 0)}{N}$$

$$F(t, q) + F(p, s) = F(t+p, q+s) + F(0, 0)$$

Here, we set $f(t, q) = F(t, q) - F(0, 0)$ because of which we have:

$$f(t, q) + f(p, s) = f(t+p, q+s)$$

The general solution of the above equations is stated as follows:

$$f(t, q) = At + Bq \Rightarrow F(t, q) = At + Bq$$

Such that $A, B, C \in \mathbb{R}$ and $F(0,0) = 0$.

Example 6.2: Let, $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that F is a continuous function and $\forall x, y, u, v \in \mathbb{R} \ \& \ N \in \mathbb{N}$ the following functional equation remains satisfied:

$$F(x - u, y - v) = F(x, y) - F(u, v)$$

To find all such applicable F .

Firstly, let $x = y = 0$ to observe the following identity:

$$F(-u, -v) = F(0,0) - F(u, v)$$

Substituting $u = -p$ and $v = -q$ such that $p, q \in \mathbb{R}$, gives us the following:

$$F(x + p, y + q) = F(x, y) - F(-p, -q)$$

By using the above established identity, we obtain the following:

$$F(x + p, y + q) = F(x, y) + F(p, q) - F(0,0)$$

Here, we say that $f(x, y) = F(x, y) - F(0,0)$, by which we have:

$$f(x + p, y + q) = f(x, y) + f(p, q)$$

The solution to the above given two-dimensional linear Cauchy equation is given as follows:

$$f(x, y) = Ax + By \Rightarrow F(x, y) = Ax + By$$

Such that $A, B \in \mathbb{R}$ and $F(0,0) = 0$.

Example 6.3: Let, $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that F is a continuous function and $\forall x, y, u, v \in \mathbb{R} \ \& \ N \in \mathbb{N}$ the following functional equation remains satisfied:

$$F(x + u, y + v) = e^{-u-v}F(x, y) + e^{-x-y}F(u, v)$$

To find all such applicable F .

Multiplying the term $e^{u+v+x+y}$ on both sides of the equation results in the following;

$$e^{u+v+x+y}F(x + u, y + v) = e^{x+y}F(x, y) + e^{u+v}F(u, v)$$

Here, we say that $f(x, y) = e^{x+y}F(x, y)$ by which we have the following:

$$f(x + u, y + v) = f(x, y) + f(u, v)$$

The general solution of the above equation is stated as follows:

$$f(x, y) = Ax + By \Rightarrow F(x, y) = e^{-x-y}(Ax + By)$$

Such that $A, B \in \mathbb{R}$.

n-dimensional Cauchy equations

In this section, we will start assuming the input dimensionality of the function f to be n . Meaning, the function f is being defined as $f : \mathbb{R}^n \rightarrow \mathbb{R}$ which happens when we assume the input variables X, Y to be n -dimensional vectors. More specifically, let us consider the following problem statement of an n -dimensional linear Cauchy equation.

Let, f be continuous function defined from $\mathbb{R}^n \rightarrow \mathbb{R}$ (Where, $n > 2$) such that $\forall X, Y \in \mathbb{R}^n$, the following equation remains satisfied.

$$f(X + Y) = f(X) + f(Y)$$

Here, $X = [x_1 \ x_2 \ \cdots \ x_n]$ and $Y = [y_1 \ y_2 \ \cdots \ y_n]$ such that $x_i, y_i \in \mathbb{R}$ for every $1 \leq i \leq n$. To find all such applicable f .

We start as per usual by extrapolating the given equation.

$$f\left(\sum_{1 \leq i \leq m} X_i\right) = \sum_{1 \leq i \leq m} f(X_i)$$

Here, $m \in \mathbb{N}_{\geq 2}$ and $X_i \in \mathbb{R}^n$ for every $1 \leq i \leq m$. Proceeding further, we set $X_i = X$ which results in the following:

$$f(mX) = mf(X)$$

Setting $X = \frac{pT}{m}$ such that $T = [t_1 \ t_2 \ \cdots \ t_n]$, $p \in \mathbb{N}$ and $T \in \mathbb{R}^n$, results in the following:

$$f(pT) = pf(T) = mf\left(\frac{pT}{m}\right)$$

$$f\left(\frac{pT}{m}\right) = \frac{p}{m}f(T)$$

This enables us to assume $q = \frac{p}{m}$ such that $q \in \mathbb{Q}$, and rewrite the above expression:

$$f(qT) = qf(T)$$

Analogous to all our previous expositions, we set a sequence of rational numbers $\{q\}$ such that its limiting behaviour is given to be $\lim_{n \rightarrow \infty} \{q_n\} = r$ such that $r \in \mathbb{R}$. Making use of this knowledge, we rewrite the above expression as follows:

$$\begin{aligned} f(q_n T) &= q_n f(T) \\ \lim_{n \rightarrow \infty} \{f(q_n T)\} &= f(rT) = rf(T) \end{aligned}$$

Now, we make sequential substitutions assuming $rt_i = s_i$ (Here, $s_i \in \mathbb{R}$) and rest every other element is assumed to be 0. By this we get a system of n equations and adding them all up results in the following:

$$f(s_1, s_2, \dots, s_n) = a_1 s_1 + a_2 s_2 + \dots + a_n s_n = \sum_{1 \leq i \leq n} a_i s_i$$

We have also assumed $f(1, 0, \dots, 0) = a_1$, $f(0, 1, \dots, 0) = a_2$ and so on until $f(0, 0, \dots, 1) = a_n$ for every $1 \leq i \leq n$ such that $a_i \in \mathbb{R}$. Therefore, our final answer can be expressed neatly using the dot product notation such that $A = [a_1 \ a_2 \ \dots \ a_n]$ and $S = [s_1 \ s_2 \ \dots \ s_n]$.

$$f(S) = A \cdot S$$

The above solution stands for every $S, A \in \mathbb{R}^n$. Using this solution, we can also solve the n -dimensional linear Pexider equation as follows:

Let, f, f_i be continuous functions defined from $\mathbb{R}^n \rightarrow \mathbb{R}$ such that $\forall X_i \in \mathbb{R}^n$ for every $1 \leq i \leq n$ and $n \in \mathbb{N}_{\geq 2}$, the following equation remains satisfied.

$$f\left(\sum_{1 \leq i \leq n} X_i\right) = \sum_{1 \leq i \leq n} f_i(X_i)$$

Such that $X_i = [x_1^{(i)} \ x_2^{(i)} \ \dots \ x_n^{(i)}]$ and $x_j^{(i)} \in \mathbb{R}$ for every $1 \leq j \leq n$. To find all such applicable f, f_i .

By making sequential substitutions, assuming all elements to be $\mathbf{0}$ except one and exhausting all possible combinations leads to a system of n equations. Adding all those equations results in the following (Note that $\mathbf{O} = [0 \ 0 \ \cdots \ 0]$):

$$\sum_{1 \leq i \leq n} f(X_i) = \sum_{1 \leq i \leq n} f_i(X_i) + (n-1) \sum_{1 \leq i \leq n} f_i(\mathbf{0})$$

$$\sum_{1 \leq i \leq n} f(X_i) = f\left(\sum_{1 \leq i \leq n} X_i\right) + (n-1)f(\mathbf{O})$$

Here, we say that $F(x_i) = f(X_i) - f(\mathbf{O})$ which leads to the following equation:

$$\sum_{1 \leq i \leq n} F(X_i) = F\left(\sum_{1 \leq i \leq n} X_i\right)$$

Solution to the above n -dimensional linear Cauchy equation is given as follows:

$$F(X) = A \cdot X \Rightarrow f(X) = A \cdot X + C$$

Such that $f(\mathbf{O}) = C$, $C \in \mathbb{R}$ and $A, X \in \mathbb{R}^n$. Using the above solution, all the subsequent general solutions are stated as follows:

$$f(X) = A \cdot X + \sum_{1 \leq i \leq n} c_i$$

$$f_i(X) = a_i x_i + c_i$$

Where, $C = \sum_{1 \leq i \leq n} c_i$ (Needless to say, $c_i \in \mathbb{R}$), $A = [a_1 \ a_2 \ \cdots \ a_n]$ and $X = [x_1 \ x_2 \ \cdots \ x_n]$ such that $a_i, x_i \in \mathbb{R}$ for every $1 \leq i \leq n$. With that done, let us now look at the n -dimensional exponential Cauchy equation.

The same generalizations can be made for other Cauchy equations as well. Let us discuss a couple of examples before proceeding to the next section.

Example 7.1: Let, f be continuous function defined from $\mathbb{R}^n \rightarrow \mathbb{R}$ (Where, $n > 2$) such that $\forall X, Y \in \mathbb{R}^n$, the following equation remains satisfied.

$$f(X - Y) = f(X) - f(Y)$$

Here, $X = [x_1 \ x_2 \ \cdots \ x_n]$ and $Y = [y_1 \ y_2 \ \cdots \ y_n]$ such that $x_i, y_i \in \mathbb{R}$ for every $1 \leq i \leq n$. To find all such applicable f .

By substituting $X = O$ such that $O = [0 \ 0 \ \cdots \ 0]$ results in the following expression:

$$f(-Y) = f(O) - f(Y)$$

Figuring out the value of $f(O)$ is quite simple and can be found out by substituting $X = Y = O$ in the original definition.

$$f(O) = 0$$

Thus, we have:

$$f(-Y) = -f(Y)$$

We assume $U = [u_1 \ u_2 \ \cdots \ u_n]$ where $u_i \in \mathbb{R}$ for every $1 \leq i \leq n$. Now substitute, $Y = -U$ in the original definition.

$$f(X + U) = f(X) + f(U)$$

As we have already solved the functional equation before, we can confidently state the final solution as follows:

$$f(X) = A \cdot X$$

Here, $A = [a_1 \ a_2 \ \cdots \ a_n]$ where $a_i \in \mathbb{R}$ for every $1 \leq i \leq n$.

Example 7.2: Let, f be continuous function defined from $\mathbb{R}^n \rightarrow \mathbb{R}$ (Where, $n > 2$) such that $\forall X, Y \in \mathbb{R}^n$ and $N \in \mathbb{N}$, the following equation remains satisfied.

$$f\left(\frac{X + Y}{N}\right) = \frac{f(X) + f(Y)}{N}$$

Here, $X = [x_1 \ x_2 \ \cdots \ x_n]$ and $Y = [y_1 \ y_2 \ \cdots \ y_n]$ such that $x_i, y_i \in \mathbb{R}$ for every $1 \leq i \leq n$. To find all such applicable f .

Substituting $X = O$ such that $O = [0 \ 0 \ \cdots \ 0]$ in the given definition:

$$f\left(\frac{Y}{N}\right) = \frac{f(O) + f(Y)}{N}$$

By substituting $X = Y = O$ in the original definition, we can easily compute the value of $f(O)$.

$$f(O) = \frac{f(O) + f(O)}{N} \rightarrow f(O) = 0$$

Returning to our original expression, we assume $Y = U + V$ where $U = [u_1 \ u_2 \ \cdots \ u_n]$ and $V = [v_1 \ v_2 \ \cdots \ v_n]$ such that $u_i, v_i \in \mathbb{R}$ for every $1 \leq i \leq n$.

$$f\left(\frac{U+V}{N}\right) = \frac{f(U+V)}{N} = \frac{f(U) + f(V)}{N}$$

$$f(U+V) = f(U) + f(V)$$

As we already know the general solution of the above equation, we state the final result as follows:

$$f(U) = A \cdot U$$

Here, $A = [a_1 \ a_2 \ \cdots \ a_n]$ where $a_i \in \mathbb{R}$ for every $1 \leq i \leq n$.

With this done, we have covered a lot of theoretical aspects of the Cauchy equations. Next up, we will be discussing deformed Cauchy equations dealing with peculiar solutions that we obtain because of over-determining the presented functional equations.

Deformed Cauchy Equations

Let us start this segment by discussing the following Cauchy equation's deformed variant.

Let, f be a continuous function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following equation remains satisfied.

$$f(x) + f(y) = f(x)f(y)$$

To find all such applicable f .

This is a classic algebraic problem and is a cornerstone for many of the Diophantine equations. We start by reorganizing the terms.

$$f(x)f(y) - f(x) = f(y)$$

Factoring the left-hand side and then dividing $f(y) - 1$ both sides.

$$f(x)(f(y) - 1) = f(y)$$

$$f(x) = \frac{f(y)}{f(y) - 1}$$

It's important to note that we implicitly assume $f(y) \neq 1$ for every $y \in \mathbb{R}$. As there is no x parameter on the right-hand side, we assume this to be c such that $c \in \mathbb{R} / \{1\}$.

$$f(x) = c$$

Substituting this back into the original functional equation results in the following:

$$2c = c^2 \rightarrow c = 0, 2$$

Thus, our final solution is stated as follows:

$$f(x) = 0, 2$$

Only constant solutions exist for f in the given conditions. Now, let us extend this deformed equation to three variables as such.

Let, f be a continuous function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y, z \in \mathbb{R}$ the following equation remains satisfied.

$$f(x) + f(y) + f(z) = f(x)f(y)f(z)$$

To find all such applicable f .

Once again, rearranging the terms and factorizing as necessary.

$$f(x)f(y)f(z) - f(x) = f(y) + f(z)$$

$$f(x)(f(y)f(z) - 1) = f(y) + f(z)$$

Dividing the term $f(y)f(z) - 1$ on both sides of the expression.

$$f(x) = \frac{f(y) + f(z)}{f(y)f(z) - 1}$$

Assuming the right-hand side to be a constant $c \in \mathbb{R}$ results in the following:

$$f(x) = c$$

Substituting the above back into the original definition results in the following:

$$3c = c^3 \rightarrow c = 0, \pm\sqrt{3}$$

Thus, our final solution being:

$$f(x) = 0, \pm\sqrt{3}$$

The same can be generalized to n variables as well and the observed result is no different from the ones we have seen so far.

Let, f be a continuous function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $x_i \in \mathbb{R}$ for every $1 \leq i \leq n$ the following equation remains satisfied.

$$\sum_{1 \leq i \leq n} f(x_i) = \prod_{1 \leq i \leq n} f(x_i)$$

To find all such applicable f .

As usual, rearranging the terms and factorizing as necessary.

$$\begin{aligned} \prod_{1 \leq i \leq n} f(x_i) - f(x_1) &= \sum_{2 \leq i \leq n} f(x_i) \\ f(x_1) \left(\prod_{2 \leq i \leq n} f(x_i) - 1 \right) &= \sum_{2 \leq i \leq n} f(x_i) \end{aligned}$$

Dividing the term $\prod_{2 \leq i \leq n} f(x_i) - 1$ on both sides.

$$f(x_1) = \frac{\sum_{2 \leq i \leq n} f(x_i)}{\prod_{2 \leq i \leq n} f(x_i) - 1}$$

Assuming the right-hand side to be a constant $c \in \mathbb{R}$. We substitute this back into the original definition to obtain the following final result (Assuming, $x \in \mathbb{R}$).

$$f(x) = c \text{ such that } c \text{ satisfies } c^n - nc = 0$$

Let us now discuss power-deformed Cauchy equations. Consider the following example.

Let, f be a continuous function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following equation remains satisfied.

$$f(x + y)^2 = f(x)^2 + f(y)^2$$

To find all such applicable f .

Firstly substituting $y = -x$ in the above given functional equation:

$$f(0) = f(x)^2 + f(-x)^2$$

But, $f(0)$ can be calculated with ease by assuming $x = y = 0$.

$$f(0)^2 = 2f(0)^2 \rightarrow f(0) = 0$$

Therefore, we have the following resulting equation:

$$f(x)^2 + f(-x)^2 = 0$$

It becomes important to realize that assuming $f(x) = a$ and $f(-x) = b$ such that $a, b \in \mathbb{R}$ results in $a^2 + b^2 = 0$. When both a and b are strictly real the given equation only holds true when $a = b = 0$. Thus, our final solution becomes the following:

$$f(x) = 0$$

But when we change our domain assumptions, the result changes drastically too. Consider the following problem statement.

Let, f be a differentiable function defined from $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that $\forall x, y \in \mathbb{R}_{\geq 0}$ the following equation remains satisfied.

$$f(x + y)^2 = f(x)^2 + f(y)^2$$

To find all such applicable f .

Taking derivative on both sides w.r.t x results in the following:

$$2f(x + y)f'(x + y) = 2f(x)f'(x)$$

Substituting $x = 0$ in the above expression assuming $2f(0)f'(0) = c$ such that $c \in \mathbb{R}_{\geq 0}$.

$$2f(y)f'(y) = c$$

$$2f(y)df(y) = c dy$$

Integrating on both sides and the integration constant $f(0) = 0$ can be ignored.

$$f(y)^2 = cy \rightarrow f(y) = \sqrt{cy}$$

It could be seen here that the solution becomes non-monotonic the moment domain assumptions are altered. To further stress on this point, let us alter the made assumptions about f as follows:

Let, f^2 have a finite and non-zero two-dimensional Laplace transform where the defined intervals are $f^2 : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ and $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that $\forall x, y \in \mathbb{R}_{\geq 0}$ the following equation remains satisfied.

$$f^2(x+y) = f^2(x) + f^2(y)$$

To find all such applicable f .

Let us firstly assume $f^2(x) = G(x)$, because of which the given functional equation changes to:

$$G(x+y) = G(x) + G(y)$$

Taking $L_{x,y}$ on both sides results in the following (Let, $L_x\{G(x)\}(p) = g(p)$, $L_y\{G(y)\}(q) = g(q)$):

$$L_{x,y}\{G(x+y)\}(p,q) = L_{x,y}\{G(x)\}(p,q) + L_{x,y}\{G(y)\}(p,q)$$

$$\frac{g(p) - g(q)}{q - p} = \frac{g(p)}{q} + \frac{g(q)}{p}$$

$$qp(g(p) - g(q)) = (q - p)(pg(p) - qg(q))$$

$$qp g(p) - qp g(q) = qp g(p) - q^2 g(q) - p^2 g(p) - pq g(q)$$

$$p^2 g(p) = q^2 g(q)$$

Assuming the right-hand side to be a constant $c \in \mathbb{R}_{\geq 0}$.

$$g(p) = \frac{c}{p^2} \rightarrow G(x) = cx$$

Therefore, our final solution is stated as such:

$$G(x) = f^2(x) = cx \rightarrow f(x) = \sqrt{cx}$$

This teaches us a very important lesson: The domain assumptions one makes on f can drastically impact its general solution. With that done, let us discuss auxiliary Cauchy equations.

Let, f, g, h, F, G, H be continuous functions defined from $f, g, h, F, G, H : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{>0}$ such that $\forall x, y \in \mathbb{R}_{\geq 0}$ the following functional equation as well as the auxiliary equation remains satisfied:

$$F(x+y)^{f(x+y)} = G(x)^{h(y)} H(y)^{g(x)}$$

With the auxiliary equation defined as:

$$f(x+y) = g(x)h(y)$$

To begin the exposition, we start by taking natural logarithms on both sides.

$$f(x+y) \ln F(x+y) = h(y) \ln G(x) + g(x) \ln H(y)$$

By using the auxiliary equation condition, we have:

$$g(x)h(y) \ln F(x+y) = h(y) \ln G(x) + g(x) \ln H(y)$$

Dividing $g(x)h(y)$ on both sides of the equation:

$$\ln F(x+y) = \frac{\ln G(x)}{g(x)} + \frac{\ln H(y)}{h(y)}$$

Here, we say that $A(x) = \ln F(x)$, $B(x) = \frac{\ln G(x)}{g(x)}$ and $C(x) = \frac{\ln H(x)}{h(x)}$. Because of which, we have the following:

$$A(x+y) = B(x) + C(y)$$

Realize that the above expression is simplified linear Pexider equation, with its solution stated down below:

$$A(x) = cx + a + b, B(x) = cx + b, C(x) = cx + a$$

Such that $a, b, c \in \mathbb{R}_{>0}$. Now, we rewrite the above solution in terms of $F(x)$, $G(x)$ and $H(x)$ instead.

$$F(x) = e^{cx+a+b}, G(x) = e^{g(x)(cx+b)}, H(x) = e^{h(x)(cx+a)}$$

Now, we shift our attention to the auxiliary equation:

$$f(x+y) = g(x)h(y)$$

The above expression is simplified exponential Pexider equation and it has the following solution:

$$f(x) = pqr^x, g(x) = pr^x, h(x) = qr^x$$

Such that $p, q, r \in \mathbb{R}_{>0}$. By using the above results, we can state our final result:

$$F(x) = e^{cx+a+b}, G(x) = e^{pr^x(cx+b)}, H(x) = e^{qr^x(cx+a)}$$

$$f(x) = pqr^x, g(x) = pr^x, h(x) = qr^x$$

Let us quickly verify the above obtained results:

$$F(x+y)^{f(x+y)} = (e^{cx+cy+a+b})^{pqr^{x+y}} = e^{pqr^{x+y}(cx+cy+a+b)}$$

$$\begin{aligned}
G(x)^{h(y)} H(y)^{g(x)} &= \left(e^{pr^x(cx+b)} \right)^{qr^y} \left(e^{qr^y(cy+a)} \right)^{pr^x} \\
&= e^{pqr^{x+y}(cx+b)} e^{pqr^{x+y}(cy+a)} = e^{pqr^{x+y}(cx+cy+a+b)}
\end{aligned}$$

Hence, our results are verified and the above stated general solution stands. Let us now discuss another variant of the above auxiliary Cauchy equation.

Let, f, g, h, F, G, H be continuous functions defined from $f, g, h, F, G, H : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$ such that $\forall x, y \in \mathbb{R}_{\geq 0}$ the following functional equation as well as the auxiliary equation remains satisfied:

$$F(x+y)f(x+y) = G(x)h(y) + H(y)g(x)$$

With the auxiliary equation given as:

$$f(x+y) = g(x)h(y)$$

To find all such applicable f, g, h, F, G, H .

To start things off, taking exponentiation on both sides of the original equation.

$$e^{F(x+y)f(x+y)} = e^{G(x)h(y)+H(y)g(x)} = e^{G(x)h(y)} e^{H(y)g(x)}$$

$$\left(e^{F(x+y)} \right)^{f(x+y)} = \left(e^{G(x)} \right)^{h(y)} \left(e^{H(y)} \right)^{g(x)}$$

Let us assume $P(x+y) = e^{F(x+y)}$, $Q(x) = e^{G(x)}$ and $R(y) = e^{H(y)}$ such that $P, Q, R : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$. Because of which we obtain:

$$P(x+y)^{f(x+y)} = Q(x)^{h(y)} R(y)^{g(x)}$$

We have already discussed the general solution of this auxiliary Cauchy equation in detail and the same is stated down below;

$$P(x) = e^{cx+a+b}, Q(x) = e^{pr^x(cx+b)}, R(x) = e^{qr^x(cx+a)}$$

$$f(x) = pqr^x, g(x) = pr^x, h(x) = qr^x$$

Such that $a, b, c, p, q, r \in \mathbb{R}_{> 0}$. Now, by using the identities $F(x) = \ln P(x)$, $G(x) = \ln Q(x)$ and $H(x) = \ln R(x)$, we state our final results:

$$F(x) = cx + a + b, G(x) = pr^x(cx + b), H(x) = qr^x(cx + a)$$

$$f(x) = pqr^x, g(x) = pr^x, h(x) = qr^x$$

Let us verify the above obtained results.

$$F(x+y)f(x+y) = (cx + cy + a + b)pqr^{x+y}$$

$$G(x)h(y) + H(y)g(x) = (cx + b)pqr^{x+y} + (cy + a)pqr^{x+y}$$

$$G(x)h(y) + H(y)g(x) = (cx + cy + a + b)pqr^{x+y}$$

The general solution obtained has been verified. With this done, let us now discuss a couple of examples before moving to the next section.

Example 8.1: Let P, Q be continuous functions defined from $P, Q : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following system of functional equations remains satisfied.

$$P(x) + P(y) = Q(x + y)$$

$$Q(x) - Q(y) = P(x - y)$$

To find all such applicable P, Q .

Firstly, substituting $x = 0$ in the first equation to obtain the following important identity:

$$P(0) + P(y) = Q(y)$$

By substituting $y = x$ in the above expression we also have:

$$P(0) + P(x) = Q(x)$$

Substituting the above two found identities in the second equation of the system.

$$(P(0) + P(x)) - (P(0) + P(y)) = P(x - y)$$

$$P(x) - P(y) = P(x - y)$$

The general solution to the above equation is given as follows (assuming, $c \in \mathbb{R}$):

$$P(x) = cx$$

Replacing the above obtained solution in the first equation of the system gives us:

$$Q(x + y) = cx + cy = c(x + y)$$

Therefore, the final general solution is stated as follows:

$$P(x) = Q(x) = cx$$

Example 8.2: Let F, G be continuous functions defined from $F, G : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following system of functional equations remains satisfied.

$$F(x + y)^2 = F(x)^2 + G(y)^2$$

$$G(x + y)^2 = G(x)^2 + F(y)^2$$

It is given that $F(0) = G(0) = 0$. To find all such applicable F, G .

Firstly, substituting $x = 0$ in the first equation of the system.

$$F(y)^2 = F(0)^2 + G(y)^2 = G(y)^2$$

Substituting the above obtained value in the second equation of the system results in the following:

$$G(x + y)^2 = G(x)^2 + G(y)^2$$

Referring to our previous expositions, we state the general solution as follows:

$$G(x) = 0$$

Now, we simply add both of the equations present in the system to obtain the following functional equation:

$$F(x + y)^2 = F(x)^2 + F(y)^2$$

Therefore, the final general solution is stated as follows:

$$F(x) = G(x) = 0$$

Let us now discuss a few more technical concepts related to functional inequalities before discussing Miscellaneous functional equations.

Convexity, Superadditivity and Subadditivity

If we were to replace the equals sign in all of our functional equations with less than or equal to or greater than or equal to signs, we would be able to connect these expressions to widely recognized functional analysis concepts. Firstly, let us define Convexity:

A continuous function f defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y, \lambda \in \mathbb{R}$ the following inequality remains satisfied then the function f is said to be convex.

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Notice that for $\lambda = \frac{1}{2}$, we obtain the following special case:

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}$$

It can be observed that by replacing lesser than or equal to sign with equals sign, we obtain a functional equation we have solved a lot previously known as the Jensen's functional equation and the above is called as Jensen's inequality.

Similarly, the definition can also be extended to n variables.

A continuous function f defined from $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $x_i, \lambda_i \in \mathbb{R}$ for every $1 \leq i \leq n$, the following inequality remains satisfied then the function f is said to be convex.

$$f\left(\sum_{1 \leq i \leq n} \lambda_i x_i\right) \leq \sum_{1 \leq i \leq n} \lambda_i f(x_i)$$

Here, $\sum_{1 \leq i \leq n} \lambda_i = 1$. There are few interesting examples that we will discuss in convex functions but before that let us familiarize ourselves with the triangle inequality. Assume, $a, b \in \mathbb{R}$ then we can formally state the following:

$$ab \leq |ab|$$

The only case where $ab = |ab|$ is when $a, b \in \mathbb{R}^+$. Otherwise, $ab < |ab|$ for any other choice of a, b . Now, as we know both sides are positive, we will be extrapolating the above expression as such:

$$2ab \leq 2|ab|$$

$$a^2 + b^2 + 2ab \leq a^2 + b^2 + 2|ab|$$

$$(|a + b|)^2 \leq (|a| + |b|)^2$$

The last step is valid as $|a||b| = |ab|$ and $(|a + b|)^2 = (a + b)^2$ for every $a, b \in \mathbb{R}$. As both sides are positive, taking a square root on both sides will still preserve the inequality.

$$|a + b| \leq |a| + |b|$$

The above expression is known as the triangle inequality. Let us generalize this statement for n variables now. Let, $a_i \in \mathbb{R}$ for every $1 \leq i \leq n$ such that $n \in \mathbb{N}_{\geq 2}$. Using the above obtained result, let us try to prove the following:

$$\left| \sum_{1 \leq i \leq n} a_i \right| \leq \sum_{1 \leq i \leq n} |a_i|$$

To start things off, let us take square on both sides:

$$\left| \sum_{1 \leq i \leq n} a_i \right|^2 \leq \left(\sum_{1 \leq i \leq n} |a_i| \right)^2$$

Using the generalized algebraic squaring formula, we can rewrite above expression as follows:

$$\sum_{1 \leq i \leq n} a_i^2 + \sum_{1 \leq i < j} \sum_{i < j \leq n} 2a_i a_j \leq \sum_{1 \leq i \leq n} a_i^2 + \sum_{1 \leq i < j} \sum_{i < j \leq n} 2|a_i a_j|$$

Cancelling out common terms on both sides, leads to the following:

$$\sum_{1 \leq i < j} \sum_{i < j \leq n} a_i a_j \leq \sum_{1 \leq i < j} \sum_{i < j \leq n} |a_i a_j|$$

The above is a true statement as for any $a_i, a_j \in \mathbb{R}$ the following statement holds true as we had discussed earlier on:

$$a_i a_j \leq |a_i a_j|$$

Hence, the double summations are simply adding such $\frac{n(n+1)}{2}$ kind of inequalities together. Therefore, the generalized triangle inequality is given as follows:

$$\left| \sum_{1 \leq i \leq n} a_i \right| \leq \sum_{1 \leq i \leq n} |a_i|$$

Similarly, one can also prove the generalized triangle inequality by using mathematical induction.

For the base step, we know that for $n = 2$ the following inequality holds for any given $a_1, a_2 \in \mathbb{R}$:

$$|a_1 + a_2| \leq |a_1| + |a_2|$$

For the assumption step, we assume that for $n = k$ such that $k \in \mathbb{N}_{\geq 2}$ the following statement holds true:

$$\left| \sum_{1 \leq i \leq k} a_i \right| \leq \sum_{1 \leq i \leq k} |a_i|$$

Such that $a_i \in \mathbb{R}$ for every $1 \leq i \leq k$. Using this assumption step, we can prove such is the case for $n = k + 1$ as well (Needless to say, $a_{k+1} \in \mathbb{R}$).

$$\left| \sum_{1 \leq i \leq k+1} a_i \right| \leq \sum_{1 \leq i \leq k+1} |a_i|$$

When we square on both sides, we obtain the following:

$$a_{k+1}^2 + 2a_{k+1} \sum_{1 \leq i \leq k} a_i + \left| \sum_{1 \leq i \leq k} a_i \right|^2 \leq a_{k+1}^2 + 2|a_{k+1}| \sum_{1 \leq i \leq k} |a_i| + \sum_{1 \leq i \leq k} |a_i|^2$$

Cancelling the common terms.

$$2a_{k+1} \sum_{1 \leq i \leq k} a_i + \left| \sum_{1 \leq i \leq k} a_i \right|^2 \leq 2|a_{k+1}| \sum_{1 \leq i \leq k} |a_i| + \sum_{1 \leq i \leq k} |a_i|^2$$

From our assumption step we already do know that $|\sum_{1 \leq i \leq k} a_i| \leq \sum_{1 \leq i \leq k} |a_i|$ which implies $|\sum_{1 \leq i \leq k} a_i|^2 \leq \sum_{1 \leq i \leq k} |a_i|^2$. Now, realize that the following identities are also known to us:

$$a_{k+1}a_1 \leq |a_{k+1}a_1|, a_{k+1}a_2 \leq |a_{k+1}a_2|, \dots, a_{k+1}a_k \leq |a_{k+1}a_k|$$

Thus, by adding them all up and multiplying 2 on both sides, we establish an important identity

$$2a_{k+1} \sum_{1 \leq i \leq k} a_i \leq 2|a_{k+1}| \sum_{1 \leq i \leq k} |a_i|$$

Henceforth, by adding the two stated key inequalities, we obtain the required result:

$$2a_{k+1} \sum_{1 \leq i \leq k} a_i + \left| \sum_{1 \leq i \leq k} a_i \right|^2 \leq 2|a_{k+1}| \sum_{1 \leq i \leq k} |a_i| + \sum_{1 \leq i \leq k} |a_i|^2$$

The above statement ultimately proves the following:

$$\left| \sum_{1 \leq i \leq k+1} a_i \right| \leq \sum_{1 \leq i \leq k+1} |a_i|$$

And that completes our induction-based generalized triangle inequality proof. Let us go back to our simplified triangle inequality using two variables and this time around, assume $a = \lambda x$ and $b = (1 - \lambda)y$ such that $x, y \in \mathbb{R}$ and $0 \leq \lambda \leq 1$, because of which we have:

$$|\lambda x + (1 - \lambda)y| \leq |\lambda x| + |(1 - \lambda)y|$$

As λ is positive, we can rewrite the above expression as such:

$$|\lambda x + (1 - \lambda)y| \leq \lambda|x| + (1 - \lambda)|y|$$

Now, if we assume $f(x) = |x|$ such that $x \in \mathbb{R}$, we can rewrite the above expression as follows:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Do realize that the above expression is simply the definition of a convex function. Hence, we state that $f(x) = |x|$ such that $x \in \mathbb{R}$ is a convex function. One may prove the same result using the generalized triangle inequality case as well.

Assume that $a_i = \lambda_i x_i$ where $1 \leq i \leq n$, $0 \leq \lambda_i \leq 1$ and $\sum_{1 \leq i \leq n} \lambda_i = 1$. Because of which we have:

$$\left| \sum_{1 \leq i \leq n} \lambda_i x_i \right| \leq \sum_{1 \leq i \leq n} |\lambda_i x_i|$$

As every $\lambda_i \geq 0$, we rewrite the above generalized expression:

$$\left| \sum_{1 \leq i \leq n} \lambda_i x_i \right| \leq \sum_{1 \leq i \leq n} \lambda_i |x_i|$$

On assuming $f(x) = |x|$ such that $x \in \mathbb{R}$, results in generalized definition of the convex function.

$$f\left(\sum_{1 \leq i \leq n} \lambda_i x_i\right) \leq \sum_{1 \leq i \leq n} \lambda_i f(x_i)$$

Hence, the conclusion remains the same, $f(x) = |x|$ for $x \in \mathbb{R}$ is a convex function. Let us now solve a couple of examples related to convex functions before discussing superadditivity and subadditivity.

Example 9.1: Suppose it is given that $f(t) = t^2$ such that $t \in \mathbb{R}$, prove that the given function is a strictly convex function.

By using the strict convex function definition, we assume $x, y \in \mathbb{R}$ and $0 < \lambda < 1$. Upon using these, we have:

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= \lambda^2 x^2 + (1 - \lambda)^2 y^2 + 2\lambda(1 - \lambda)xy \\ \lambda f(x) + (1 - \lambda)f(y) &= \lambda x^2 + (1 - \lambda)y^2 \end{aligned}$$

By substituting above obtained computations in our definition, we have the following:

$$\lambda^2 x^2 + (1 - \lambda)^2 y^2 + 2\lambda(1 - \lambda)xy < \lambda x^2 + (1 - \lambda)y^2$$

Subtracting the term $\lambda x^2 + (1 - \lambda)y^2$ on both sides results in the following:

$$\begin{aligned} (\lambda^2 x^2 - \lambda x^2) + ((1 - \lambda)^2 y^2 - (1 - \lambda)y^2) + 2\lambda(1 - \lambda)xy &< 0 \\ \lambda(\lambda - 1)x^2 + (1 - \lambda)(-\lambda)y^2 + 2\lambda(1 - \lambda)xy &< 0 \\ \lambda(\lambda - 1)x^2 + \lambda(\lambda - 1)y^2 - 2\lambda(\lambda - 1)xy &< 0 \end{aligned}$$

Dividing $\lambda(\lambda - 1)$ will flip the sign of the inequality as $0 < \lambda < 1$ and $-1 < 1 - \lambda < 0$ which implies $\lambda(1 - \lambda) < 0$.

$$\begin{aligned} x^2 + y^2 - 2xy &\geq 0 \\ (x - y)^2 &\geq 0 \end{aligned}$$

We know the above inequality to be true as square of any number cannot be less than zero. Therefore, $f(t) = t^2$ where $t \in \mathbb{R}$ is a strictly convex function.

Example 9.2: Suppose it is given that $f(x, y) = |x + y|$ such that $x, y \in \mathbb{R}^2$. Prove that the given function is a convex function.

By using the standard convex function definition, we assume $\forall x_1, x_2, y_1, y_2 \in \mathbb{R}$ and $0 \leq \lambda \leq 1$ to compute the following:

$$\begin{aligned} f(\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)) &= |\lambda(x_1 + y_1) + (1 - \lambda)(x_2 + y_2)| \\ \lambda f(x_1, y_1) + (1 - \lambda)f(x_2, y_2) &= \lambda|x_1 + y_1| + (1 - \lambda)|x_2 + y_2| \end{aligned}$$

Upon setting $X = x_1 + y_1$ and $Y = x_2 + y_2$, we obtain the following weighted triangle inequality:

$$|\lambda X + (1 - \lambda)Y| \leq \lambda|X| + (1 - \lambda)|Y|$$

We have already proven the above statement to be true in our previous expositions. Therefore, $f(x, y) = |x + y|$ such that $x, y \in \mathbb{R}^2$ is a convex function.

Example 9.3: Suppose the following inequality holds where $x_i \in \mathbb{R}$ for every $1 \leq i \leq N$ given $N \in \mathbb{N}_{\geq 2}$ and f is a continuous function defined from $f : \mathbb{R} \rightarrow \mathbb{R}$.

$$f\left(\frac{1}{N} \sum_{1 \leq i \leq N} x_i\right) \leq \frac{1}{N} \sum_{1 \leq i \leq N} f(x_i)$$

Find out the nature of function f .

The given inequality is a direct consequence of the convexity definition. Upon setting $\lambda_i = \frac{1}{N}$ for every $1 \leq i \leq N$, we have the following:

$$f\left(\sum_{1 \leq i \leq N} \lambda_i x_i\right) \leq \sum_{1 \leq i \leq N} \lambda_i f(x_i)$$

Do notice that the other required conditions such as $\sum_{1 \leq i \leq N} \lambda_i = 1$ and $0 \leq \lambda_i \leq 1$ for every $1 \leq i \leq N$ are also essentially true. Thus, as the above expression is the n variable definition of convexity, we can confidently state that f is convex by nature.

With that done, we move forward and discuss superadditivity and subadditivity.

If, for a given continuous function F defined from $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x, y \in \mathbb{R}$ the following inequality holds true:

$$F(x) + F(y) \leq F(x + y)$$

Then, the function F is said to be superadditive. Similarly, if the signs are reversed:

$$F(x) + F(y) \geq F(x + y)$$

Then, the function F is said to be subadditive. Similarly, if one wants to introduce weights and correlate the above definitions with convexity, then we introduce $0 \leq \lambda \leq 1$ as follows:

$$F(\lambda x) + F((1 - \lambda)y) \leq F(\lambda x + (1 - \lambda)y)$$

$$F(\lambda x) + F((1 - \lambda)y) \geq F(\lambda x + (1 - \lambda)y)$$

The above given function F is superadditive and subadditive respectively. From the original definition itself, one could see that $F(t) = |t|$ where $t \in \mathbb{R}$ is both a convex function as well as a subadditive function. Assume $x, y \in \mathbb{R}$ then we observe the following:

$$F(x) + F(y) = |x| + |y|$$

$$F(x + y) = |x + y|$$

From our previous discussions, we already know the following triangle inequality:

$$|x| + |y| \geq |x + y|$$

Therefore, we have the following final result:

$$F(x) + F(y) \geq F(x + y)$$

Hence, $F(t) = |t|$ where $t \in \mathbb{R}$ is subadditive. Coming back to the structural reason as to why F is both convex as well as subadditive is because of the following property (Assume $0 \leq \lambda \leq 1$):

$$|\lambda x| = \lambda|x|, |(1 - \lambda)y| = (1 - \lambda)|y|$$

Which essentially implies:

$$F(\lambda x) = \lambda F(x), F((1 - \lambda)y) = (1 - \lambda)F(y)$$

Because of which, both of the required inequalities will hold. Let us now cover a few examples before discussing the last chapter of this book.

Example 9.4: Suppose it is given that $f(x) = \sqrt{x}$ where $x \in \mathbb{R}_{\geq 0}$, figure out the nature of f in terms of subadditivity, superadditivity and convexity.

Firstly, we will be testing the convexity of the given function using the weight parameter λ such that $0 \leq \lambda \leq 1$.

$$f(\lambda x + (1 - \lambda)y) = \sqrt{\lambda x + (1 - \lambda)y}$$

$$\lambda f(x) + (1 - \lambda)f(y) = \lambda\sqrt{x} + (1 - \lambda)\sqrt{y}$$

By substituting them in the Jensen's inequality, we obtain the following:

$$\sqrt{\lambda x + (1 - \lambda)y} \leq \lambda\sqrt{x} + (1 - \lambda)\sqrt{y}$$

Upon setting $x = 1, y = 0$, we observe the following:

$$\sqrt{\lambda} \leq \lambda$$

Now, recall that $0 \leq \lambda \leq 1$ because of which when we take square root over λ it will always be greater than or equal to λ . The edge cases of equality occur when $\lambda = 0, 1$ so actually the correct statement would be the opposite $\sqrt{\lambda} \geq \lambda$. Because of this contradiction, f is not convex.

Let us now proceed with the subadditivity test, by using the standard definition (It is assumed that $x, y \in \mathbb{R}_{\geq 0}$).

$$f(x) + f(y) = \sqrt{x} + \sqrt{y}$$

$$f(x + y) = \sqrt{x + y}$$

Substituting them in the definition, taking square on both sides and then cancelling the common terms results in the following:

$$\sqrt{x} + \sqrt{y} \geq \sqrt{x + y} \rightarrow x + y + 2\sqrt{xy} \geq x + y$$

$$\sqrt{xy} \geq 0$$

But as $x, y \geq \mathbb{R}_{\geq 0}$, the above inequality holds and because of which we can say that f is subadditive. No need to check superadditivity as the observed inequality will be $\sqrt{xy} \leq 0$ which is obviously false. Hence, f is neither convex or superadditive but subadditive by nature.

Example 9.5: Suppose, it is given that the function $f(x)$ satisfies the following inequality given that $x \in \mathbb{R}_{\geq 0}$:

$$f(x) + f(y) \leq f(x + y)$$

To generate a template using which n many solutions of f can be extracted (where $n \in \mathbb{N}$).

Let us start with the solution obtained from linear Cauchy equation which does satisfy the given inequality as an edge case (Assuming, $c > 0$).

$$f(x) = cx \rightarrow f(x) + f(y) = f(x + y)$$

Similarly, let us now change the function value to $f(x) = c\sqrt{x}$ instead and see whether it satisfies the required inequality.

$$c(\sqrt{x} + \sqrt{y}) \geq c\sqrt{x+y} \rightarrow \sqrt{x} + \sqrt{y} \geq \sqrt{x+y}$$

The above statement is true as observed in the previous example itself. Now changing the function value to $f(x) = cx^{\frac{1}{3}}$ instead.

$$x^{\frac{1}{3}} + y^{\frac{1}{3}} \geq (x+y)^{\frac{1}{3}}$$

Taking cube on both sides results in the following:

$$x + y + 3x^{\frac{2}{3}}y^{\frac{1}{3}} + 3x^{\frac{1}{3}}y^{\frac{2}{3}} \geq x + y \rightarrow x^{\frac{2}{3}}y^{\frac{1}{3}} + x^{\frac{1}{3}}y^{\frac{2}{3}} \geq 0$$

The above inequality has to be true as both $x, y \in \mathbb{R}_{\geq 0}$. Similarly, for a general case say $n \in \mathbb{N}_{\geq 2}$, the function value is set to be $f(x) = cx^{\frac{1}{n}}$.

$$x^{\frac{1}{n}} + y^{\frac{1}{n}} \geq (x+y)^{\frac{1}{n}} \rightarrow \left(x^{\frac{1}{n}} + y^{\frac{1}{n}}\right)^n \geq (x+y)$$

$$x + y + \sum_{1 \leq i \leq n-1} \binom{n}{i} x^{\frac{i}{n}} y^{\frac{n-i}{n}} \geq x + y$$

By subtracting $x + y$ on both sides we obtain the following:

$$\sum_{1 \leq i \leq n} \binom{n}{i} x^{\frac{i}{n}} y^{\frac{n-i}{n}} \geq 0$$

This inequality must be true as $x, y \in \mathbb{R}_{\geq 0}$ and the left-hand side simply acts as a positive residue of the performed expansion. Hence, the general template f can be described as follows:

$$f(x) = cx^{\frac{1}{n}}$$

Such that $n \in \mathbb{N}$ and $c > 0$. With this done, we conclude our discussion of the theory of Cauchy equations. If the reader is interested, I'd highly recommend

reading about non-continuous Cauchy equations and Hamel-basis solutions. It requires a good foundation in set theory but for well-versed enthusiasts it would prove to be a good read.

Appendix

Laplace transform in one-dimension is stated as follows:

$$L_x\{f(x)\}(p) = \int_0^{\infty} e^{-px} f(x) dx = F(p)$$

Few important properties of Laplace transform in one-dimension are listed as follows assuming $L_x\{g(x)\}(p) = G(p)$ and $a > 0$:

- (i) $L_x\{cf(x)\}(p) = cF(p)$ (such that $c \in \mathbb{R}$)
- (ii) $L_x\{bg(x) \pm cf(x)\}(p) = bG(p) \pm cF(p)$ (such that $b, c \in \mathbb{R}$)
- (iii) $L_x\{f(ax)\}(p) = \frac{1}{a} F\left(\frac{p}{a}\right)$
- (iv) $L_x\{e^{ax}f(x)\}(p) = F(p - a)$

Along with that, few important key results are also listed down below assuming $a > 0$:

- (i) $L_x\{c\}(p) = \frac{c}{p}$ (such that $c \in \mathbb{R}$)
- (ii) $L_x\{x^n\}(p) = \frac{n!}{p^{n+1}}$ (such that $n \in \mathbb{N}$)
- (iii) $L_x\{e^{ax}\}(p) = \frac{1}{p-a}$
- (iv) $L_x\{\sin(ax)\}(p) = \frac{a}{p^2+a^2}$
- (v) $L_x\{\cos(ax)\}(p) = \frac{p}{p^2+a^2}$

Moving forward, we now discuss the double-sided Laplace transform.

$$L_x^*\{f(x)\}(p) = \int_{\mathbb{R}} e^{-px} f(x) dx = \tilde{f}(p)$$

Few important properties of double-sided Laplace transforms are stated as follows assuming $L_x^*\{g(x)\}(p) = \tilde{g}(p)$ and $a > 0$:

- (i) $L_x^*\{cf(x)\}(p) = c\tilde{f}(p)$ (such that, $c \in \mathbb{R}$)
- (ii) $L_x^*\{bg(x) \pm cf(x)\}(p) = b\tilde{g}(p) \pm c\tilde{f}(p)$ (such that, $b, c \in \mathbb{R}$)

- (iii) $L_x^*\{f(x+a)\}(p) = e^{ap}\tilde{f}(p)$
- (iv) $L_x^*\{f(ax)\}(p) = \frac{1}{a}\tilde{f}\left(\frac{p}{a}\right)$
- (v) $L_x^*\{e^{ax}f(x)\}(p) = \tilde{f}(p-a)$

Extending further, we now discuss two-dimensional Laplace transform.

$$L_{x,y}\{f(x,y)\}(p,q) = \int_0^\infty \int_0^\infty e^{-px-qy} f(x,y) dx dy = \bar{F}(p,q)$$

Few important properties of two-dimensional Laplace transforms are listed down below assuming $L_{x,y}\{g(x,y)\}(p,q) = \bar{G}(p,q)$ and $\alpha, \beta > 0$:

- (i) $L_{x,y}\{cf(x,y)\}(p,q) = cF(p,q)$ (such that, $c \in \mathbb{R}$)
- (ii) $L_{x,y}\{cf(x,y) \pm bg(xy)\}(p,q) = c\bar{F}(p,q) \pm b\bar{G}(p,q)$ (such that, $b, c \in \mathbb{R}$)
- (iii) $L_{x,y}\{f(\alpha x, \beta y)\}(p,q) = \frac{1}{\alpha\beta} \bar{F}\left(\frac{p}{\alpha}, \frac{q}{\beta}\right)$
- (iv) $L_{x,y}\{e^{\alpha x + \beta y} f(x,y)\}(p,q) = \bar{F}(p-\alpha, q-\beta)$
- (v) $L_{x,y}\{f(x+y)\}(p,q) = \frac{F(p)-F(q)}{q-p}$ (Where, $L_x\{f(x)\}(p) = F(p)$ and $L_y\{f(y)\}(q) = F(q)$)
- (vi) When f is odd: $L_{x,y}\{f(x-y)\}(p,q) = \frac{F(p)-F(q)}{p+q}$ (Where, $L_x\{f(x)\}(p) = F(p)$ and $L_y\{f(y)\}(q) = F(q)$)
- (vii) When f is even: $L_{x,y}\{f(x-y)\}(p,q) = \frac{F(p)+F(q)}{p+q}$ (Where, $L_x\{f(x)\}(p) = F(p)$ and $L_y\{f(y)\}(q) = F(q)$)

Key results can be easily deduced from the one-dimensional Laplace transform itself and as an example $L_{x,y}\{\sin(ax+by)\}(p,q)$ is calculated down below assuming $a, b > 0$:

$$\begin{aligned} L_{x,y}\{\sin(ax+by)\}(p,q) &= \int_0^\infty \int_0^\infty e^{-px-qy} \sin(ax+by) dx dy \\ &= \int_0^\infty \int_0^\infty e^{-px-qy} (\sin(ax) \cos(by) + \cos(ax) \sin(by)) dx dy \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} e^{-qy} \cos(by) dy \int_0^{\infty} e^{-px} \sin(ax) dx + \int_0^{\infty} e^{-px} \cos(ax) dx \int_0^{\infty} e^{-qy} \sin(by) dy \\
&= \left(\frac{p}{p^2 + b^2} \right) \left(\frac{a}{p^2 + a^2} \right) + \left(\frac{p}{p^2 + a^2} \right) \left(\frac{b}{p^2 + b^2} \right) = \frac{(a + b)p}{(p^2 + b^2)(p^2 + a^2)}
\end{aligned}$$

Therefore, our final solution is given as follows:

$$L_{x,y}\{\sin(ax + by)\}(p, q) = \frac{(a + b)p}{(p^2 + b^2)(p^2 + a^2)}$$

Moving forward, let us now discuss about the Mellin transforms.

$$M_x\{f(x)\}(s) = \int_0^{\infty} x^{s-1} f(x) dx = \psi(s)$$

Few important properties of Mellin transforms are listed down below assuming $M_x\{g(x)\}(s) = \phi(s)$ and $a > 0$:

- (i) $M_x\{cf(x)\}(s) = c\psi(s)$ (such that, $c \in \mathbb{R}$)
- (ii) $M_x\{bf(x) \pm cg(x)\}(s) = b\psi(s) + c\phi(s)$ (such that, $b, c \in \mathbb{R}$)
- (iii) $M_x\{f(ax)\}(s) = a^{-s}\psi(s)$
- (iv) $M_x\{x^c f(x)\}(s) = \psi(s + c)$ (such that, $c \in \mathbb{R}$)
- (v) $M_x\{x^a f(x)\}(s - a) = \psi(s)$

Along with that, couple of important results are also stated as follows assuming $\delta(x)$ denotes standard Dirac-delta function, $\Gamma(p)$ denotes the gamma function and $a > 0$:

- (i) $M_x\{\delta(x)\}(p) = A$ (Where, δ is the standard Dirac-delta function symbol)
- (ii) $M_x\{\delta(x - a)\}(p) = a^{s-1}$
- (iii) $M_x\{e^{-x}\}(p) = \Gamma(p)$

With this, we wrap up the appendix section, using the above given identities and formulae will certainly assist a lot in solving functional equations from the chapter *Transformable Cauchy equations*.

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Mathematics often hides great depth behind simple expressions.
The Theory of Cauchy functional Equations stands as one of the
Finest examples of this beauty.